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A Physics-Derived Loss-Aware Linear Mixing Model for Hyperspectral Unmixing with Spectral Variability

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***Abstract*—Hyperspectral unmixing (HU) aims to decompose mixed pixels into endmembers and their corresponding abundances. However, spectral variability arising from intrinsic material properties and environmental conditions introduces scaling variability that the classical linear mixing model (LMM) fails to adequately characterize. To address this, we extend the LMM by proposing a novel physics-derived framework named the Loss-Aware Linear Mixing Model (LALMM). Grounded in an absorbing Markov chain, LALMM explicitly simulates the light propagation process by introducing propagation loss and capture probability, and derives a pixel-wise scaling factor from the limiting transition matrix of the Markov chain. This formulation provides a rigorous physical interpretation of multiplicative spectral variability as a consequence of energy dissipation during light propagation, establishing a principled connection between radiative transfer physics and the scaling coefficients commonly adopted in LMM variants. To solve LALMM efficiently, we further develop the Loss-Aware Nonnegative Tensor Factorization (LA-NTF) algorithm, which employs the core tensor as a structured low-rank representation of the spectral variability effects modeled by LALMM and incorporates joint spatial–spectral constraints to regularize the enlarged solution space associated with the additional latent variables. Experimental results on both synthetic and real-world datasets demonstrate the superior unmixing accuracy and robustness of the proposed LA-NTF compared to several state-of-the-art algorithms.**

***Index Terms*—Hyperspectral image (HSI), spectral variability, nonnegative tensor factorization (NTF), unsupervised spectral unmixing, Markov chain.**

I. Introduction

HYPERSPECTRAL images (HSI), owing to their rich spectral information, have been widely applied across numerous fields [1] [2]. However, the high spectral resolution inevitably comes at the cost of reduced spatial resolution. Furthermore, due to atmospheric effects and intimate mixing, mixed pixels are pervasive in HSI, which severely hinders downstream applications. To address this issue, hyperspectral unmixing (HU) has emerged as an effective solution that decomposes mixed pixels into a set of pure spectral signatures (endmembers), along with their corresponding fractional proportions (abundances) [3].

HU methods are primarily grounded in two assumptions: the linear mixing model (LMM) and the nonlinear mixing model (NLMM) [4]. The LMM assumes that the observed spectrum can be expressed as a linear combination of the products of endmembers and their corresponding abundances, plus additive noise. However, this simplified formulation limits the model's capacity to capture complex spectral variations, making it inadequate for handling multiplicative noise, spectral variability [5], and other effects that are prevalent in complex scenes. In contrast, NLMM such as the bilinear mixing model (BMM) [6] [7], generalized bilinear model (GBM) [8] and the multilinear mixing model (MLM) [9] employ polynomial terms to characterize the nonlinear interactions between pixels and endmembers [10], thereby offering superior physical interpretability in scenes with complex spectral interactions.

Therefore, optimizing the LMM to enhance its spectral variability representation capability has become an important problem. To address this, Veganzones et al. [11] [12] proposed the extended linear mixing model (ELMM), which directly incorporates pixel-wise scaling factors into the LMM to account for spectral variability. Thouvenin et al. [13] proposed the perturbed linear mixing model (PLMM), which models spectral variability through an additive perturbation term. Hong et al. [14] proposed the augmented LMM (ALMM), which combines the ELMM and PLMM by jointly employing multiplicative and additive factors to characterize spectral variability. Furthermore, Imbiriba et al. [15] proposed the generalized linear mixing model (GLMM), which extends the scalar scaling factor in the ELMM to a band-wise formulation, further enhancing its spectral variability fitting capability. Recently, spectral variability augmented unmixing methods, including sparse unmixing and deep learning based frameworks, have also been developed to further improve the modeling of endmember variability [16] [17] [18]. In particular, Gao et al. [19] proposed a spatial–spectral adaptive fusion network (SSAF-Net) that uses variational generative modeling to account for endmember variability.

To accurately solve the established spectral mixing models, a large number of effective algorithms have been proposed. Based on the LMM, endmember extraction algorithms such as vertex component analysis (VCA) [20], N-FINDR [21], pixel purity index (PPI) [22], and minimum volume simplex analysis (MVSA) [23] [24] have been developed. When

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endmembers are known, the fully constrained least squares (FCLS) [25] algorithm is commonly employed to estimate abundances. Subsequently, nonnegative matrix factorization (NMF) [26] [27] was introduced to simultaneously estimate both endmembers and abundances, offering intuitive interpretation and clear physical significance.

To mitigate the non-convexity of the NMF objective function, reduce sensitivity to initialization, and avoid convergence to local optima, various regularization terms have been incorporated into the NMF framework, including sparsity regularization [28], spectral similarity constraints [29], spatial similarity constraints [30], and minimum volume constraints [31]. To further enhance the performance of NMF-based methods, Qin et al. [32] introduced graph total variation regularization into the NMF framework (gtvMBO) and adopted the Merriman–Bence–Osher scheme to solve the objective function in a bitwise manner. Chen et al. [33] proposed an adaptive multi-order graph regularized NMF method (MOGNMF), which combines multi-order graph regularization with dual sparsity constraints to improve unmixing robustness. Zhang et al. [34] proposed EISNMF, which integrates a nonsmooth matrix and constrained energy minimization based spatial sparse regularization into the NMF framework to improve endmember and abundance estimation. With the advancement of deep learning, Xiong et al. [35] proposed a novel unrolling-based hyperspectral unmixing method (SNMF-Net), and Cui et al. [36] proposed group sparsity regularized unmixing unrolling (GSUU), both of which integrate data-driven learning paradigms to further improve the performance and generalization capability of NMF-based unmixing methods.

Although these NMF-based algorithms have demonstrated promising performance, their inherent formulation requires the conversion of three-dimensional hyperspectral images into two-dimensional matrices, which inevitably leads to information loss. As an alternative, tensors [37] can be regarded as a natural representation of HSI [38], where an HSI patch comprising two spatial dimensions and one spectral dimension can be represented by a third-order tensor. Tensors have been widely applied in HSI classification [39] [40], anomaly detection [41] [42], and image reconstruction [43]. In HU, tensor-based methods have also attracted increasing attention. Zhang et al. [44] first introduced nonnegative tensor factorization (NTF) to the unmixing task. Subsequently, Qian et al. [45] proposed matrix–vector NTF (MV-NTF), which elegantly integrates the LMM with block-term decomposition (BTD) to provide a physically interpretable unmixing framework. Xiong et al. [46] extended MV-NTF by incorporating total variation (TV) regularization to promote spatial smoothness in the estimated abundance maps. Wang et al. [47] proposed endmember independence constrained HU via NTF (EIC-NTF), introducing an endmember independence constraint and a low-rank constraint into the NTF framework. Li et al. [48] proposed SCNMTF, which combines NTF with NMF and incorporates an $L_{1/2}$ regularization term. Gao et al. [49] applied NTF to the GBM and imposed a low-rank constraint on the nonlinear component. Qu et al. [50] proposed MFUTV-LRNTF, which introduces a horizontal unidirectional TV (UTV) operator to jointly regularize all three factor matrices in the NTF. Recently, Xiong [51] proposed overparameterized nonnegative tensor factorization (ONTF), which employs nested nonlinear factorizations and implicit regularization to capture complex low-rank structures in abundance maps.

Although methods such as the ELMM and the GLMM have achieved notable progress in mitigating spectral variability, their core mechanisms rely on intuition-driven assumptions that lack rigorous physical justification and mathematical derivation. In real remote sensing imaging scenarios, spectral variability is not merely a simple numerical scaling effect, but rather the combined result of energy attenuation, path scattering, and absorption effects experienced by photons propagating through complex media [52]. Furthermore, existing NTF-based unmixing algorithms commonly adopt the BTD framework, and to maintain physical consistency with LMM, the core tensor in the BTD is often simplified to an identity matrix. This simplification assumes that the spatial basis vector components are mutually independent with no cross-coupling interactions. However, since real-world ground cover textures are often irregular and ground cover distributions are spatially intermingled, such an assumption introduces certain limitations in practical geographic scenarios. Retaining the core tensor enables the modeling of more complex spectral variations and interactions among spatial basis vectors, allowing the model to capture more generalizable spatial topological structures. Nevertheless, how to integrate a BTD with a retained core tensor into a spectral mixing model while maintaining physical consistency remains an open question.

To address the above issues, we propose a novel physics-derived spectral mixing model termed the Loss-Aware LMM (LALMM), and further develop an NTF-based hyperspectral unmixing algorithm (LA-NTF) upon this foundation. The main contributions are summarized as follows:

1) To address the lack of physical grounding in existing scaling-based LMM variants, LALMM is grounded in an absorbing Markov chain framework, introducing a propagation loss q and a capture probability p to simulate energy attenuation during photon transmission. LALMM rigorously derives a pixel-wise scaling factor, establishing an explicit connection between radiative transfer physics and the scaling coefficients adopted in existing LMM variants.
2) To bridge the physically derived LALMM and the NTF framework, we develop LA-NTF, which departs from conventional NTF by retaining and optimizing the core tensor in the $(L, L, 1)$-BTD instead of fixing it to an identity tensor. Together with the associated spatial factors, the retained core tensor forms a structured low-rank representation of the spatially varying scaling effects modeled by LALMM, thereby extending conventional NTF to accommodate LALMM-induced spectral variability.
3) To regularize the enlarged solution space introduced by the loss-aware scaling representation, LA-NTF incorporates TV and MV constraints to promote spatial smoothness of abundance maps and geometric compactness of endmembers, alleviating the ill-posedness caused by the additional variability-related variables.

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The remainder of this paper is organized as follows. Section II briefly reviews the preliminaries of the LMM and tensor decomposition. Section III explains the detail of the proposed LALMM and the LA-NTF algorithm. Section IV analyzes the experimental results on both synthetic and real-world datasets and compares them with several state-of-the-art algorithms. Section V concludes this article.

II. Related Work

In this section, we first introduce the Linear Mixing Model (LMM) and then we review several fundamental concepts of tensor decomposition.

A. LMM

Given an HSI $\mathbf{Y} \in \mathbb{R}^{B\times N}$ with B spectral bands and $N = H \times W$ pixels, where k denotes the number of endmembers, the ith mixed pixel $\mathbf{y}_i \in \mathbb{R}^{B\times 1}$ can be formulated in the LMM as

$$\mathbf{y}_i = \mathbf{E}\mathbf{a}_i + \mathbf{n}_i, \tag{1}$$

where $\mathbf{E} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_k) \in \mathbb{R}^{B\times k}$ is the endmember matrix, $\mathbf{a}_i = (a_{i,1}, a_{i,2}, \ldots, a_{i,k})^T \in \mathbb{R}^{k\times 1}$ is the abundance vector associated with $\mathbf{y}_i$, and $\mathbf{n}_i \in \mathbb{R}^{B\times 1}$ denotes the corresponding noise.

The abundance vector $\mathbf{a}_i$ is constrained by the abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC).

$$\begin{cases} \forall j, a_{i,j} \geq 0, & \text{(ANC)} \\ \sum_{j=1}^{k} a_{i,j} = 1, & \text{(ASC)}. \end{cases} \tag{2}$$

Extending (1) to the entire HSI yields

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{N}, \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{k\times N}$ is the abundance matrix and $\mathbf{N} \in \mathbb{R}^{B\times N}$ represents the noise.

B. Tensor Decomposition Operations and Concepts

A tensor is a multidimensional array [37]. An nth-order tensor has n modes, where each mode represents a distinct dimension of the tensor. In the following, we introduce the tensor decomposition operations and concepts employed in this paper.

Definition of Matricization: For an n-order tensor $\mathbf{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_n}$, the mode-$i$ matricization rearranges the mode-i vectors into columns of a matrix. The resulting matrix is denoted as $\mathbf{X}_{(i)} \in \mathbb{R}^{I_i\times\prod_{j\neq i} I_j}$.

Definition of Kronecker Product: Given two matrices $\mathbf{A} \in \mathbb{R}^{I\times J}$ and $\mathbf{B} \in \mathbb{R}^{K\times L}$, their Kronecker product, denoted by $\mathbf{A}\otimes\mathbf{B} \in \mathbb{R}^{IK\times JL}$, is defined as

$$\mathbf{A}\otimes\mathbf{B} = \begin{pmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1J}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2J}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{I1}\mathbf{B} & a_{I2}\mathbf{B} & \cdots & a_{IJ}\mathbf{B} \end{pmatrix}. \tag{4}$$

Definition of Khatri-Rao Product: The Khatri-Rao product of two matrices $\mathbf{A} \in \mathbb{R}^{I\times J}$ and $\mathbf{B} \in \mathbb{R}^{K\times J}$ with the same number of columns J, denoted by $\mathbf{A}\odot\mathbf{B} \in \mathbb{R}^{IK\times J}$, is defined as

$$\mathbf{A}\odot\mathbf{B} = (\mathbf{a}_1\otimes\mathbf{b}_1, \mathbf{a}_2\otimes\mathbf{b}_2, \cdots, \mathbf{a}_J\otimes\mathbf{b}_J), \tag{5}$$

where $\mathbf{a}_j$ and $\mathbf{b}_j$ denote the jth columns of $\mathbf{A}$ and $\mathbf{B}$, respectively.

Definition of k-Mode Product: The k-mode product of a tensor $\mathbf{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_K}$ with a matrix $\mathbf{A} \in \mathbb{R}^{J\times I_k}$, denoted by $\mathbf{X}\times_k\mathbf{A}$, results in a tensor of size $I_1\times\cdots\times I_{k-1}\times J\times I_{k+1}\times\cdots\times I_K$. The elements of the resulting tensor are given by

$$(\mathbf{X}\times_k\mathbf{A})_{i_1\cdots i_{k-1}ji_{k+1}\cdots i_K} = \sum_{i_k=1}^{I_k} x_{i_1i_2\cdots i_{k-1}i_ki_{k+1}\cdots i_K} a_{ji_k}. \tag{6}$$

Definition of Outer Product: The outer product of two tensors $\mathbf{A} \in \mathbb{R}^{I_1\times I_2\times\cdots I_P}$ and $\mathbf{B} \in \mathbb{R}^{J_1\times J_2\times\cdots J_Q}$ is the tensor $\mathbf{A}\circ\mathbf{B} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_P\times J_1\times J_2\times\cdots\times J_Q}$ and its elements are defined by

$$(\mathbf{A}\circ\mathbf{B})_{i_1i_2\cdots i_Pj_1j_2\cdots j_Q} = a_{i_1i_2\ldots i_P}b_{j_1j_2\ldots j_Q}. \tag{7}$$

Definition of Generalized Khatri–Rao Product: The generalized Khatri–Rao product of two tensors is denoted by $\bar{\odot}$. It is defined for partitioned matrices that share the same number of submatrices. Specifically, let $\mathbf{A}$ and $\mathbf{B}$ be two matrices partitioned into k submatrices, i.e., $\mathbf{A} = (\mathbf{A}_1, \cdots, \mathbf{A}_k)$ and $\mathbf{B} = (\mathbf{B}_1, \cdots, \mathbf{B}_k)$. Their generalized Khatri–Rao product is defined as

$$\mathbf{A}\bar{\odot}\mathbf{B} = (\mathbf{A}_1\otimes\mathbf{B}_1, \cdots, \mathbf{A}_k\otimes\mathbf{B}_k). \tag{8}$$

Definition of Block Term Decomposition: Given an nth-order tensor $\mathcal{T} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_n}$, BTD represents it as a sum of D block terms. Each block term is obtained by performing the k-mode product between a core tensor $\mathcal{G}_d \in \mathbb{R}^{R_1\times R_2\times\cdots\times R_n}$ and n factor matrices $\mathbf{A}_d^{(k)} \in \mathbb{R}^{I_k\times R_k}$ along the kth mode of $\mathcal{G}_d$, where $d \in \{1, \ldots, D\}$ and $k \in \{1, \ldots, n\}$. The BTD model is formulated as

$$\mathcal{T} = \sum_{d=1}^{D} \mathcal{G}_d \times_1 \mathbf{A}_d^{(1)} \times_2 \mathbf{A}_d^{(2)} \times_3 \cdots \times_n \mathbf{A}_d^{(n)}. \tag{9}$$

In HU, BTD is often adopted in the rank-$(L, L, 1)$ form [45] [46], which can be expressed as

$$\mathcal{T} = \sum_{d=1}^{D} \mathcal{G}_d \times_1 \mathbf{A}_d \times_2 \mathbf{B}_d \times_3 \mathbf{c}_d = \sum_{d=1}^{D} \left(\mathbf{A}_d \cdot \mathcal{G}_d \cdot \mathbf{B}_d^T\right) \circ \mathbf{c}_d, \tag{10}$$

where $\mathbf{c}_d$ is a vector along the third mode.

III. Proposed Method

In this section, we first introduce the proposed spectral mixing model and then present the unsupervised unmixing algorithm developed based on it.

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A. LALMM

LMM has been widely adopted in hyperspectral unmixing due to its simplicity and clear physical interpretability. However, classical LMM neglects the inevitable energy loss that occurs during light propagation, which leads to spectral variability and ultimately limits its ability to accurately describe spectral mixing processes in complex scenes. To address this limitation, we incorporate the energy loss during light propagation into the spectral mixing process and formulate light propagation as an absorbing Markov chain. Based on this idea, we propose a new spectral mixing model, termed Loss-Aware Linear Mixing Model (LALMM). LALMM not only extends the LMM but also provides a novel interpretation of the spectral variability coefficients appearing in several commonly used variants of LMM, revealing their physical meaning from the perspective of light propagation.

The losses occurring during light propagation, including but not limited to atmospheric attenuation [53] and energy loss during reflection, are collectively referred to as propagation losses in this paper. In practical unmixing methods, such losses are typically treated as model error and absorbed into the noise term. However, studies on radiative transfer theory [54] and rendering equations [55] have demonstrated that propagation losses are non-negligible and difficult to simplify. Moreover, studies on spectral variability [5] [52] have further shown that atmospheric effects and environmental factors tend to induce significant spectral variability. Therefore, explicitly incorporating propagation losses into spectral mixing models has become a meaningful and pressing task.

We model the propagation losses in a probabilistic manner. Specifically, after each interaction with a material, a photon associated with the ith pixel transitions to one of the following three states:

- The photon is lost during propagation with probability q_i, where $q_i \in [0, 1)$.
- The photon is directly captured by the sensor with probability p_i, where $p_i \in (0, 1], (p_i + q_i) \in (0, 1]$.
- With the remaining probability $(1-p_i-q_i)$, the photon is reflected and interacts with another material. In this case, the interaction probabilities among different materials are characterized by their abundance fractions, following the assumption of the multilinear spectral mixing model [9].

Formally, let $S_t \in \{1, 2, \cdots, k, k+1, k+2\}$ denote the material state at the tth interaction, where states 1 to k correspond to the k endmembers, state $k+1$ represents propagation losses, and state $k+2$ corresponds to photon captured by the sensor. Given the current state $S_t = h$, the transition probability to the next state $S_{t+1} = j$ is defined as

$$P\left(S_{t+1}=j|S_t=h\right)=\begin{cases}(1-p_i-q_i)a_{i,j}, & j=1,2,\ldots,k,\\ q_i, & j=k+1,\\ p_i, & j=k+2,\end{cases} \tag{11}$$

where $a_{i,j}$ denotes the abundance of the jth endmember in the ith pixel. Here, p_i and q_i are interpreted as effective propagation parameters at the pixel level, which characterize the average attenuation and sensor capture behavior associated with repeated interactions between photons and ground covers within a mixed pixel. When the number of endmembers is two, the reflection process is illustrated in Fig. 1.

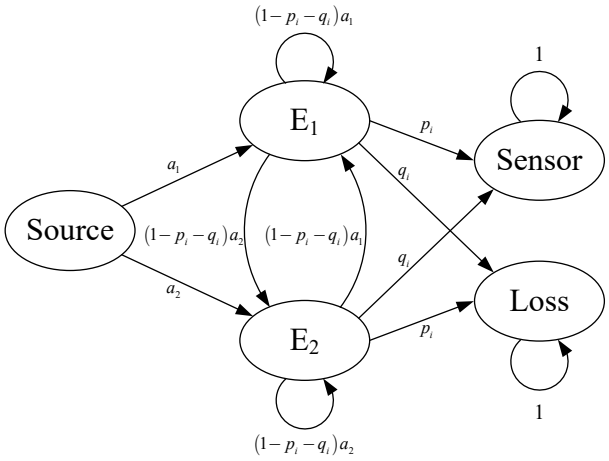


Fig. 1. Illustration of the proposed absorbing Markov chain model for light propagation among the source, two endmembers, sensor, and loss states.

Therefore, the state distribution of photons after multiple reflections among materials can be modeled by a Markov state transition matrix. The initial state distribution for the ith pixel is defined as $\pi_{i,0} = \begin{pmatrix} a_{i,1} & a_{i,2} & \cdots & a_{i,k} & 0 & 0 \end{pmatrix}$, and the one-step transition probabilities are summarized in Table I.

Table I can be equivalently represented by a transition matrix $\mathbf{M}_i \in \mathbb{R}^{(k+2)\times(k+2)}$, given by

$$\mathbf{M}_i = \begin{pmatrix} \mathbf{Q}_i & \mathbf{R}_i \\ \mathbf{0} & \mathbf{I}_2 \end{pmatrix} \tag{12}$$

where

$$\begin{aligned} \mathbf{Q}_i &= (1-p_i-q_i)\begin{pmatrix} a_{i,1} & a_{i,2} & \cdots & a_{i,k} \\ a_{i,1} & a_{i,2} & \cdots & a_{i,k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,1} & a_{i,2} & \cdots & a_{i,k} \end{pmatrix}, \\ \mathbf{R}_i &= \begin{pmatrix} q_i & p_i \\ q_i & p_i \\ \vdots & \vdots \\ q_i & p_i \end{pmatrix}, \mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \end{aligned} \tag{13}$$

where $\mathbf{Q}_i \in \mathbb{R}^{k\times k}$ represents the transition probabilities among endmembers, $\mathbf{R}_i \in \mathbb{R}^{k\times 2}$ corresponds to the transition probabilities from endmembers to the two absorbing states, and $\mathbf{I}_2$ denotes a 2×2 identity matrix. After n reflections, the transition matrix can be expressed as $\mathbf{M}_i^n$.

We aim to characterize the state distribution when all photons are eventually absorbed by the system, which amounts to computing the asymptotic state distribution. Specifically, we first analyze the convergence of $\mathbf{M}_i^n$. According to the theory of absorbing Markov chains [56], $\lim_{n\to\infty} \mathbf{M}_i^n$ always converges to a limiting matrix, which can be represented as

$$\lim_{n\to\infty} \mathbf{M}_i^n = \begin{pmatrix} \mathbf{0}_{k\times k} & \frac{1}{p_i+q_i}\mathbf{R}_i \\ \mathbf{0}_{2\times k} & \mathbf{I}_2 \end{pmatrix}. \tag{14}$$

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TABLE I
ONE-STEP STATE TRANSITION PROBABILITIES

State	EM#1	EM#2	$\cdots$	EM#k	Propagation Loss	Sensor Capture
EM#1	$(1-p_i-q_i)a_{i,1}$	$(1-p_i-q_i)a_{i,2}$	$\cdots$	$(1-p_i-q_i)a_{i,k}$	q_i	p_i
EM#2	$(1-p_i-q_i)a_{i,1}$	$(1-p_i-q_i)a_{i,2}$	$\cdots$	$(1-p_i-q_i)a_{i,k}$	q_i	p_i
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
EM#k	$(1-p_i-q_i)a_{i,1}$	$(1-p_i-q_i)a_{i,2}$	$\cdots$	$(1-p_i-q_i)a_{i,k}$	q_i	p_i
Propagation Loss	0	0	$\cdots$	0	1	0
Sensor Capture	0	0	$\cdots$	0	0	1

The detailed derivation of (14) is provided in the Appendix, where $\mathbf{0}_{i\times j}$ denotes an $i\times j$ zero matrix. Therefore, the limiting distribution $\lim_{n\to\infty}\pi_{i,n}$ can be expressed as

$$\begin{aligned}\lim_{n\to\infty}\pi_{i,n} &= \lim_{n\to\infty}\pi_{i,0}\cdot\mathbf{M}_i^n \\ &= \begin{pmatrix} a_{i,1} & a_{i,2} & \cdots & a_{i,k} & 0 & 0 \end{pmatrix}\cdot\begin{pmatrix} \mathbf{0}_{k\times k} & \frac{1}{p_i+q_i}\mathbf{R}_i \\ \mathbf{0}_{2\times k} & \mathbf{I}_2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & \cdots & 0 & \frac{q_i}{p_i+q_i}\sum_{j=1}^{k}a_{i,j} & \frac{p_i}{p_i+q_i}\sum_{j=1}^{k}a_{i,j} \end{pmatrix} \\ &= \begin{pmatrix} 0 & \cdots & 0 & \frac{q_i}{p_i+q_i} & \frac{p_i}{p_i+q_i} \end{pmatrix}.\end{aligned} \tag{15}$$

In HU, the observed spectrum is determined by the fraction of photons that are eventually captured by the sensor. According to (15), for a photon whose initial state lies in the endmember set $j\in\{1,2,\cdots,k\}$, the probability that it is ultimately detected by the sensor, denoted by α_i for the ith pixel, is given by

$$\begin{gathered}\alpha_i = \lim_{n\to\infty}\mathbf{M}_i^n(j,k+2) = \frac{p_i}{p_i+q_i}. \\ p_i>0, q_i\geq 0, 0<p_i+q_i\leq 1\end{gathered} \tag{16}$$

The final observed spectrum at the ith pixel can be expressed as

$$\begin{gathered}\mathbf{y}_i = \alpha_i\sum_{j=1}^{k}a_{i,j}\mathbf{e}_j+\mathbf{n}_i = \alpha_i\mathbf{E}\mathbf{a}_i+\mathbf{n}_i, \\ \text{s.t. } \alpha_i\in(0,1].\end{gathered} \tag{17}$$

where $\mathbf{e}_j,\mathbf{a}_i,\mathbf{E}$ and $\mathbf{n}_i$ are as previously defined in (1). In particular, when $q_i=0$, the propagation loss is absent, and the model degenerates to the standard LMM.

In addition to the absorbing Markov chain formulation, we provide an alternative derivation of the proposed spectral mixing model. We define the event B_m as the photon associated with the ith pixel being captured by the sensor at the end of the mth interaction, where $m\in\{1,2,3,\cdots\}$. The probability of B_m is given by

$$\Pr(B_m) = p_i(1-p_i-q_i)^{m-1}. \tag{18}$$

When the mth interaction occurs with a certain material, the corresponding material distribution at the ith pixel is given by $\mathbf{a}_i$. Therefore, the expected observed spectrum $\mathbf{y}_i^m$ at the mth step is given by

$$\mathbf{y}_i^m = \Pr(B_m)\cdot\mathbf{E}\mathbf{a}_i = p_i(1-p_i-q_i)^{m-1}\mathbf{E}\mathbf{a}_i. \tag{19}$$

Based on (18) (19), the total reflected spectrum $\mathbf{y}_i$ can be expressed as

$$\begin{aligned}\mathbf{y}_i &= \sum_{m=1}^{\infty}\mathbf{y}_i^m = \sum_{m=1}^{\infty}p_i(1-p_i-q_i)^{m-1}\mathbf{E}\mathbf{a}_i+\mathbf{n}_i \\ &= \frac{p_i}{p_i+q_i}\mathbf{E}\mathbf{a}_i+\mathbf{n}_i = \alpha_i\mathbf{E}\mathbf{a}_i+\mathbf{n}_i, \\ &\qquad \text{s.t. } \alpha_i\in(0,1].\end{aligned} \tag{20}$$

We then extend the formulation to the entire hyperspectral image, and the resulting spectral mixing model can be expressed as

$$\mathbf{Y} = \mathbf{E}(\mathbf{A}\Gamma)+\mathbf{N}, \tag{21}$$

where $\Gamma=\mathrm{diag}(\alpha_1,\alpha_2,\ldots,\alpha_N)\in\mathbb{R}^{N\times N}$.

According to the transition matrix $\mathbf{M}_i$, the expected number of steps required to reach the absorbing state can be computed as

$$\mathbf{T}_i = (\mathbf{I}_k-\mathbf{Q}_i)^{-1}\mathbf{1} = \frac{1}{p_i+q_i}\mathbf{1}, \tag{22}$$

where $\mathbf{1}\in\mathbb{R}^{k\times 1}$ denotes an all-ones vector. Since all entries of $\mathbf{T}_i$ are identical, we denote their common value by $T_i=\frac{1}{p_i+q_i}$. As $p_i+q_i\in(0,1]$, we have $T_i\in[1,\infty)$. This indicates that a photon must undergo at least one interaction before reaching an absorbing state, which is consistent with physical intuition.

It should be noted that the current LALMM mainly describes attenuation-dominated spectral variability, while signal enhancement caused by atmospheric scattering or other gain effects is not explicitly modeled.

B. LA-NTF

In this subsection, we propose Loss-Aware NTF (LA-NTF), an algorithm designed to solve the LALMM. The flowchart of LA-NTF is illustrated in Fig. 2. LALMM introduces a diagonal matrix $\Gamma\in\mathbb{R}^{N\times N}$, which acts as a spatially varying per-pixel scaling on the reconstructed HSI. Although Γ is diagonal, directly estimating it remains computationally challenging due to the massive number of parameters involved ($N=H\times W$). While deep learning approaches may implicitly model such spatial modulation using convolutional or fully connected layers, this strategy is not suitable for traditional unmixing methods. Therefore, reducing the dimensionality of Γ prior to its estimation is necessary.

First, Γ can be incorporated into the tensor factorization model through a structured core tensor representation. Specifically, the spatial variation of Γ can be projected into a latent

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subspace and parameterized within the BTD framework. In this way, the unconstrained estimation of pixel-wise scaling coefficients is replaced by a structured low-rank representation, while maintaining structural consistency with the BTD formulation in (10).

From the perspective of BTD, the LALMM can be expressed as

$$\begin{aligned}\mathcal{X} &= \sum_{i=1}^{k}(\Gamma_{H\times W}.*\mathbf{A}_i)\circ\mathbf{e}_i+\mathcal{N}\\ &\approx \sum_{i=1}^{k}\mathcal{G}_L\times_1\mathbf{B}_i\times_2\mathbf{C}_i\times_3\mathbf{e}_i+\mathcal{N}\\ &\approx \sum_{i=1}^{k}\mathbf{B}_i\cdot\mathcal{G}_L\cdot{\mathbf{C}_i}^T\circ\mathbf{e}_i+\mathcal{N},\end{aligned} \tag{23}$$

where $\mathcal{X},\mathcal{N}\in\mathbb{R}^{H\times W\times B}$ denote third-order tensors. $\mathcal{X}$ is the tensorized form of $\mathbf{Y}$; $\Gamma_{H\times W}\in\mathbb{R}^{H\times W}$ is the spatial rearrangement of the diagonal elements of Γ; $\mathcal{G}_L\in\mathbb{R}^{L\times L\times 1}$ denotes the core tensor. The operator $.*$ represents the Hadamard product. Here, $\mathbf{B}_i\in\mathbb{R}^{H\times L}$ and $\mathbf{C}_i\in\mathbb{R}^{W\times L}$ are spatial bases, and $\mathbf{A}_i=\mathbf{B}_i\mathbf{C}_i^T\in\mathbb{R}^{H\times W}$ is the abundance map of the ith endmember.

Instead of directly estimating Γ, we represent its spatial variation using a structured low-rank parameterization based on the core tensor $\mathcal{G}_L\in\mathbb{R}^{L\times L\times 1}$. Specifically, $\mathcal{G}_L$ is shared across spectral bands and lifted to the original spatial domain by the bases $\mathbf{B}_i$ and $\mathbf{C}_i$, thereby providing an efficient approximation of the element-wise spatial scaling. This construction replaces the unconstrained estimation of pixel-wise scaling coefficients with a low-rank structured representation, which regularizes the spatial variability of Γ and improves the stability of the optimization. For notational convenience, we identify $\mathcal{G}_L$ with its frontal slice $\mathrm{diag}(g_1,g_2,\ldots,g_L)$.

Ignoring the noise term $\mathcal{N}$, $\mathcal{X}$ can be expressed in matrix form as

$$\begin{aligned}\mathbf{X}_{(1)}&=\left[(\mathbf{C}_1\odot\mathbf{B}_1)\,\mathcal{G}_L\mathbf{1}_L\cdots(\mathbf{C}_k\odot\mathbf{B}_k)\,\mathcal{G}_L\mathbf{1}_L\right]\mathbf{E}^T,\\ \mathbf{X}_{(2)}&=(\mathbf{C}\bar{\odot}\mathbf{E})\,\mathcal{G}_{Lk}\mathbf{B}^T,\\ \mathbf{X}_{(3)}&=(\mathbf{E}\bar{\odot}\mathbf{B})\,\mathcal{G}_{Lk}\mathbf{C}^T,\\ \mathcal{G}_{Lk}&=\mathbf{I}_k\otimes\mathcal{G}_L,\end{aligned} \tag{24}$$

where $\mathbf{X}_{(1)}\in\mathbb{R}^{HW\times B}$, $\mathbf{X}_{(2)}\in\mathbb{R}^{WB\times H}$, and $\mathbf{X}_{(3)}\in\mathbb{R}^{BH\times W}$ are the mode-1, mode-2, and mode-3 unfoldings of $\mathcal{X}$, respectively. Here, $\mathbf{B}=[\mathbf{B}_1,\ldots,\mathbf{B}_k]\in\mathbb{R}^{H\times Lk}$; $\mathbf{C}=[\mathbf{C}_1,\ldots,\mathbf{C}_k]\in\mathbb{R}^{W\times Lk}$; $\mathbf{E}=[\mathbf{e}_1,\ldots,\mathbf{e}_k]\in\mathbb{R}^{B\times k}$. The vector $\mathbf{1}_L\in\mathbb{R}^L$ is an all-ones vector; $\mathcal{G}_{Lk}\in\mathbb{R}^{Lk\times Lk}$ is a block-diagonal matrix; and $\mathbf{I}_k$ denotes the $k\times k$ identity matrix.

To solve for the unknown parameters in (23), we minimize the reconstruction error (RE) within a maximum likelihood framework. Specifically, the objective function is expressed as:

$$\begin{aligned}\text{minimize}\quad & f(\mathbf{B},\mathbf{C},\mathcal{G}_L,\mathbf{E})=\frac{1}{2}\left\|\mathbf{X}_{(3)}-(\mathbf{E}\bar{\odot}\mathbf{B})\mathcal{G}_{Lk}\mathbf{C}^T\right\|_F^2\\ \text{subject to}\quad & \mathbf{B},\mathbf{C},\mathbf{E}\geq 0;\quad \mathbf{B}\mathbf{C}^T=\mathbf{1}_{H\times W};\\ & g_j>0,\forall j=1,\ldots,L,\end{aligned} \tag{25}$$

where $\mathbf{1}_{H\times W}$ is an $H\times W$ all-ones matrix.

Subsequently, we incorporate constraint terms into the objective function to improve the optimization performance and impose physically meaningful constraints. First, to ensure the physical interpretability of the abundances, the abundance sum-to-one constraint is imposed, which is defined as

$$J_{\text{ASC}}(\mathbf{B},\mathbf{C})=\frac{\lambda_0}{2}\left\|\mathbf{1}_{H\times W}-\mathbf{B}\mathbf{C}^T\right\|_F^2, \tag{26}$$

where $\lambda_0\in\mathbb{R}$ is the regularization parameter.

Based on the spatial smoothness prior of hyperspectral images, we introduce a total variation (TV) regularization term on the abundances. By exploiting spatial correlations, TV preserves edges while enforcing spatial smoothness and enhances the robustness of the unmixing results. TV term can be expressed as

$$J_{\text{TV}}(\mathbf{B},\mathbf{C})=\frac{\lambda_1}{2}\text{TV}\left(\mathbf{B}\mathbf{C}^T\right)=\frac{\lambda_1}{2}\sum_{i=1}^{N}\sqrt{\sum_{j\in\mathcal{N}(i)}w_{ij}(a_i-a_j)^2}, \tag{27}$$

where $\lambda_1\in\mathbb{R}$ is the regularization parameter, $\text{TV}(\cdot)$ denotes the total variation operator, $\mathbf{W}$ is a weighted adjacency matrix, and w_{ij} represents the similarity weight between pixel i and j. $\mathcal{N}(i)$ denotes the set of neighboring pixels of pixel i, and an 8-connected neighborhood system is adopted in this work.

To further enhance the physical interpretability of the extracted endmembers and mitigate geometric ambiguity, we incorporate a minimum volume (MV) constraint into the model. This constraint encourages the estimated endmembers to form a simplex with minimum volume, thereby improving the robustness and accuracy of the unmixing process. The MV constraint term is formulated as

$$\begin{aligned}&J_{\text{MV}}(\mathbf{E})=\\ &\frac{\lambda_2}{2}\det{}^2\left(\begin{bmatrix}\mathbf{1}_k^T\\ \underline{\mathbf{0}}\end{bmatrix}+\begin{bmatrix}\mathbf{0}_{k-1}^T\\ \underline{\mathbf{I}}\end{bmatrix}\mathbf{H}^T\left(\mathbf{E}-\widetilde{\mathbf{H}}\mathbf{1}_k^T\right)\right),\end{aligned} \tag{28}$$

where $\lambda_2\in\mathbb{R}$ is the regularization parameter, $\mathbf{1}_k^T\in\mathbb{R}^k$ denotes the all-ones row vector, $\underline{\mathbf{0}}\in\mathbb{R}^{(k-1)\times k}$ is a $(k-1)\times k$ all-zeros matrix, $\mathbf{0}_{k-1}^T$ is an all-zeros vector, $\underline{\mathbf{I}}\in\mathbb{R}^{(k-1)\times k}$ is a $(k-1)\times k$ truncated identity matrix. $\mathbf{H}\in\mathbb{R}^{B\times(k-1)}$ is obtained by applying PCA to the observed HSI $\mathbf{Y}$, and consists of the $(k-1)$ principal components. The vector $\widetilde{\mathbf{H}}\in\mathbb{R}^B$ denotes the mean vector of the pixels in $\mathbf{Y}$.

The objective function with constraints can be formulated as

$$\begin{aligned}f(\mathbf{B},\mathbf{C},\mathcal{G}_L,\mathbf{E})=&\frac{1}{2}\left\|\mathbf{X}_{(3)}-(\mathbf{E}\bar{\odot}\mathbf{B})\mathcal{G}_{Lk}\mathbf{C}^T\right\|_F^2+\\ &J_{\text{ASC}}(\mathbf{B},\mathbf{C})+J_{\text{TV}}(\mathbf{B},\mathbf{C})+J_{\text{MV}}(\mathbf{E}).\end{aligned} \tag{29}$$

To improve convergence stability, we introduce auxiliary variables $\mathbf{U}=\mathbf{B}$ and $\mathbf{V}=\mathbf{C}$. The resulting objective function

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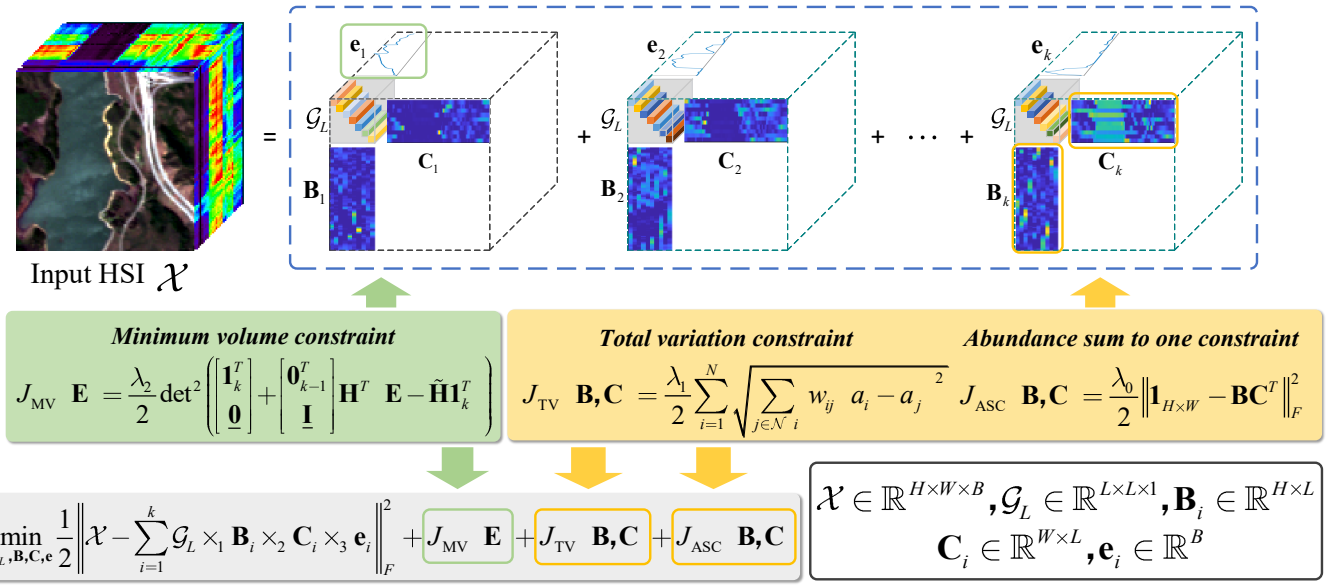


Fig. 2. Overview of the proposed LA-NTF framework. The input HSI tensor $\mathcal{X}$ is approximated by the sum of k material components, where k is the number of endmembers. For component i, $\mathbf{e}_i$ denotes the endmember spectrum, and $\mathbf{B}_i$ and $\mathbf{C}_i$ are nonnegative spatial bases defining the abundance map $\mathbf{A}_i = \mathbf{B}_i\mathbf{C}_i^T$ with rank at most L. The shared core tensor $\mathcal{G}_L$ provides a structured approximation of the loss-induced spatial scaling in LALMM. Each component is reconstructed through mode products $\times_n$ of the core tensor and the corresponding factors. Alternating optimization of the bottom objective estimates the endmember spectra, spatial bases, and core tensor, with abundance maps recovered from the spatial bases. The regularization terms J_{MV}, J_{TV}, and J_{ASC} promote endmember simplex compactness, spatial smoothness of abundances, and a unit abundance sum per pixel, respectively.

is reformulated as

$$\begin{aligned} f(\mathbf{B},\mathbf{C},\mathcal{G}_L,\mathbf{E},\mathbf{U},\mathbf{V}) &= \frac{1}{2}\left\|\mathbf{X}_{(3)}-(\mathbf{E}\bar{\odot}\mathbf{B})\mathcal{G}_{Lk}\mathbf{C}^T\right\|_F^2 \\ &+\frac{\lambda_0}{2}\left\|\mathbf{1}_{H\times W}-\mathbf{B}\mathbf{C}^T\right\|_F^2+\frac{\lambda_1}{2}\mathrm{TV}\left(\mathbf{U}\mathbf{V}^T\right) \\ &+\frac{\lambda_2}{2}\det{}^2\left(\begin{bmatrix}\mathbf{1}_k^T\\ \underline{\mathbf{0}}\end{bmatrix}+\begin{bmatrix}\mathbf{0}_{k-1}^T\\ \underline{\mathbf{I}}\end{bmatrix}\mathbf{H}^T\left(\mathbf{E}-\tilde{\mathbf{H}}\mathbf{1}_k^T\right)\right) \\ &+\frac{\mu}{2}\left(\|\mathbf{U}-\mathbf{B}\|_F^2+\|\mathbf{V}-\mathbf{C}\|_F^2\right), \end{aligned} \tag{30}$$

where $\mu>0$ is the penalty parameter controlling the equality constraints $\mathbf{U}=\mathbf{B}$ and $\mathbf{V}=\mathbf{C}$.

C. Optimization Algorithm

We employ the alternating optimization framework to minimize the objective function in (30). At each iteration, we update one variable while keeping the others fixed. The detailed update rules for the $(t+1)$th iteration are derived as follows.

1) Update for $\mathbf{E}$*:* The subproblem for $\mathbf{E}$ is formulated as

$$\begin{aligned} \mathbf{E}^{(t+1)} &= \underset{\mathbf{E}}{\operatorname{argmin}}\ \frac{1}{2}\left\|\mathbf{X}_{(1)}-\mathbf{M}_1^{(t)}\mathbf{E}^T\right\|_F^2 \\ &+\frac{\lambda_2}{2}\det{}^2\left(\begin{bmatrix}\mathbf{1}_k^T\\ \underline{\mathbf{0}}\end{bmatrix}+\begin{bmatrix}\mathbf{0}_{k-1}^T\\ \underline{\mathbf{I}}\end{bmatrix}\mathbf{H}^T\left(\mathbf{E}-\tilde{\mathbf{H}}\mathbf{1}_k^T\right)\right), \end{aligned} \tag{31}$$

where we define

$$\mathbf{M}_1 = \left[(\mathbf{C}_1\odot\mathbf{B}_1)\,\mathcal{G}_L\mathbf{1}_L\cdots(\mathbf{C}_k\odot\mathbf{B}_k)\,\mathcal{G}_L\mathbf{1}_L\right]. \tag{32}$$

The multiplicative update rule for (31) is given by

$$\mathbf{E}^{(t+1)} = \mathbf{E}^{(t)}.*\frac{\mathbf{X}_{(1)}^T\mathbf{M}_1^{(t)}}{\mathbf{E}^{(t)}\left(\mathbf{M}_1^{(t)}\right)^T\mathbf{M}_1^{(t)}+\lambda_2\left(\mathbf{H}^T\mathbf{H}\right)\mathbf{E}^{(t)}}, \tag{33}$$

where $.*$ denotes the Hadamard product, and the fraction denotes element-wise division.

2) Update for $\mathbf{B}$*:* The subproblem for $\mathbf{B}$ is formulated as

$$\begin{aligned} \mathbf{B}^{(t+1)} &= \underset{\mathbf{B}}{\operatorname{argmin}}\ \frac{1}{2}\left\|\mathbf{X}_{(2)}-\mathbf{M}_2^{(t)}\mathbf{B}^T\right\|_F^2+ \\ &\frac{\lambda_0}{2}\left\|\mathbf{1}_{H\times W}-\mathbf{B}\left(\mathbf{C}^{(t)}\right)^T\right\|_F^2+\frac{\mu}{2}\left\|\mathbf{U}^{(t)}-\mathbf{B}\right\|_F^2. \end{aligned} \tag{34}$$

Let

$$\mathbf{M}_2 = (\mathbf{C}\bar{\odot}\mathbf{E})\,\mathcal{G}_{Lk}. \tag{35}$$

The multiplicative update rule for (34) is given by

$$\mathbf{B}^{(t+1)} = \mathbf{B}^{(t)}.*\frac{\mathbf{X}_{(2)}^T\mathbf{M}_2^{(t)}+\lambda_0\mathbf{1}_{H\times W}\mathbf{C}^{(t)}+\mu\mathbf{U}^{(t)}}{\mathbf{B}^{(t)}\left(\mathbf{M}_2^{(t)}\right)^T\mathbf{M}_2^{(t)}+\lambda_0\mathbf{B}^{(t)}\left(\mathbf{C}^{(t)}\right)^T\mathbf{C}^{(t)}+\mu\mathbf{B}^{(t)}}. \tag{36}$$

3) Update for $\mathbf{C}$*:* The subproblem for $\mathbf{C}$ is formulated as

$$\begin{aligned} \mathbf{C}^{(t+1)} &= \underset{\mathbf{C}}{\operatorname{argmin}}\ \frac{1}{2}\left\|\mathbf{X}_{(3)}-\mathbf{M}_3^{(t)}\mathbf{C}^T\right\|_F^2+ \\ &\frac{\lambda_0}{2}\left\|\mathbf{1}_{H\times W}-\mathbf{B}^{(t+1)}\mathbf{C}^T\right\|_F^2+\frac{\mu}{2}\left\|\mathbf{V}^{(t)}-\mathbf{C}\right\|_F^2, \end{aligned} \tag{37}$$

where we define

$$\mathbf{M}_3 = (\mathbf{E}\bar{\odot}\mathbf{B})\,\mathcal{G}_{Lk}. \tag{38}$$

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The multiplicative update rule for (37) is given by

$$\mathbf{C}^{(t+1)} = \mathbf{C}^{(t)} .* \frac{\mathbf{X}_{(3)}^T \mathbf{M}_3^{(t)} + \lambda_0 \mathbf{1}_{H\times W}^T \mathbf{B}^{(t+1)} + \mu \mathbf{V}^{(t)}}{\mathbf{C}^{(t)} \left(\mathbf{M}_3^{(t)}\right)^T \mathbf{M}_3^{(t)} + \lambda_0 \mathbf{C}^{(t)} \left(\mathbf{B}^{(t+1)}\right)^T \mathbf{B}^{(t+1)} + \mu \mathbf{C}^{(t)}}. \tag{39}$$

4) Update for $\mathbf{U}$*:* The subproblem for $\mathbf{U}$ is formulated as

$$\mathbf{U}^{(t+1)} = \underset{\mathbf{U}}{\operatorname{argmin}} \frac{\mu}{2} \left\| \mathbf{U} - \mathbf{B}^{(t+1)} \right\|_F^2 + \frac{\lambda_1}{2} \mathrm{TV} \left(\mathbf{U} \left(\mathbf{V}^{(t)} \right)^T \right). \tag{40}$$

The multiplicative update rule for (40) is given by

$$\mathbf{U}^{(t+1)} = \mathbf{U}^{(t)} .* \frac{\mu \mathbf{B}^{(t+1)} + \lambda_1 \mathbf{W} \mathbf{U}^{(t)}}{\mu \mathbf{U}^{(t)} + \lambda_1 \mathbf{D} \mathbf{U}^{(t)}}, \tag{41}$$

where $\mathbf{D}$ is a diagonal matrix whose entries are the row sums of $\mathbf{W}$.

5) Update for $\mathbf{V}$*:* The subproblem for $\mathbf{V}$ is formulated as

$$\mathbf{V}^{(t+1)} = \underset{\mathbf{V}}{\operatorname{argmin}} \frac{\mu}{2} \left\| \mathbf{V} - \mathbf{C}^{(t+1)} \right\|_F^2 + \frac{\lambda_1}{2} \mathrm{TV} \left(\mathbf{U}^{(t+1)} \mathbf{V}^T \right). \tag{42}$$

The multiplicative update rule for (42) is given by

$$\mathbf{V}^{(t+1)} = \mathbf{V}^{(t)} .* \frac{\mu \mathbf{C}^{(t+1)} + \lambda_1 \mathbf{W} \mathbf{V}^{(t)}}{\mu \mathbf{V}^{(t)} + \lambda_1 \mathbf{D} \mathbf{V}^{(t)}}. \tag{43}$$

6) Update for $\mathcal{G}_L$*:* The subproblem for $\mathcal{G}_L$ is formulated as

$$\mathcal{G}_L^{(t+1)} = \underset{\mathcal{G}_L}{\operatorname{argmin}} \frac{1}{2} \left\| \mathbf{X}_{(3)} - (\mathbf{E} \bar{\odot} \mathbf{B}) \mathcal{G}_{Lk} \mathbf{C}^T \right\|_F^2. \tag{44}$$

We define $\mathbf{M}_4 = \mathbf{E} \bar{\odot} \mathbf{B}$ and $\mathbf{P} = \mathbf{1}_k \otimes \mathbf{I}_L$, $\mathbf{1}_k \in \mathbb{R}^k$ denotes an all-ones column vector, $\mathbf{I}_L$ denotes the $L \times L$ identity matrix. The multiplicative update rule for $\mathcal{G}_{Lk}$ is given by

$$\mathcal{G}_{Lk}^{(t+1)} = \mathcal{G}_{Lk}^{(t)} .* \operatorname{diag}\left(\frac{\mathbf{P}^T \left(\mathbf{M}_4^{(t+1)}\right)^T \mathbf{X}_{(3)} \mathbf{C}^{(t+1)} \mathbf{1}_{Lk}}{\mathbf{P}^T \left(\mathbf{M}_4^{(t+1)}\right)^T \mathbf{M}_4^{(t+1)} \mathcal{G}_{Lk}^{(t)} \left(\mathbf{C}^{(t+1)}\right)^T \mathbf{C}^{(t+1)} \mathbf{P} \mathcal{G}_{Lk}^{(t)} \mathbf{1}_{Lk}} \right). \tag{45}$$

Then $\mathcal{G}_L^{(t+1)}$ can be expressed as

$$\mathcal{G}_L^{(t+1)} = \frac{1}{k} \sum_{i=1}^{L} \left(u_i u_i^T \cdot \operatorname{tr} \left((u_i \otimes \mathbf{1}_k)^T \mathcal{G}_{Lk}^{(t+1)} (u_i \otimes \mathbf{1}_k) \right) \right), \tag{46}$$

where u_i denotes the standard basis vector.

The entire algorithm is presented in Algorithm 1.

Algorithm 1: Pseudocode of the proposed method

Input: Original HSI $\mathcal{X}$, Endmember number k, Tensor Rank L

Output: Endmembers $\mathbf{E}$, Abundances $\mathbf{A}$, Coefficient Matrix $\mathcal{G}_L$

Random initialize $\mathbf{E}$ and $\mathcal{G}_L$.

Initialize $\mathbf{A}$ using FCLS.

Set $[\mathbf{B}, \mathbf{C}] = \mathrm{SVD}(\mathbf{A}), \mathbf{B} = \max(\mathbf{B}, \epsilon), \mathbf{C} = \max(\mathbf{C}, \epsilon), \mathbf{U} = \mathbf{B}, \mathbf{V} = \mathbf{C}$.

for $t < T_{max}$ **do**

  $\mathbf{E}^{(t+1)} \leftarrow \mathbf{E}^{(t)} .* \frac{\mathbf{X}_{(1)}^T \mathbf{M}_1^{(t)}}{\mathbf{E}^{(t)} \left(\mathbf{M}_1^{(t)}\right)^T \mathbf{M}_1^{(t)} + \lambda_2 (\mathbf{H}^T \mathbf{H}) \mathbf{E}^{(t)}}$.

  $\mathbf{B}^{(t+1)} \leftarrow \mathbf{B}^{(t)} .* \frac{\mathbf{X}_{(2)}^T \mathbf{M}_2^{(t)} + \lambda_0 \mathbf{1}_{H\times W} \mathbf{C}^{(t)} + \mu \mathbf{U}^{(t)}}{\mathbf{B}^{(t)} \left(\mathbf{M}_2^{(t)}\right)^T \mathbf{M}_2^{(t)} + \lambda_0 \mathbf{B}^{(t)} \left(\mathbf{C}^{(t)}\right)^T \mathbf{C}^{(t)} + \mu \mathbf{B}^{(t)}}$.

  $\mathbf{C}^{(t+1)} \leftarrow \mathbf{C}^{(t)} .* \frac{\mathbf{X}_{(3)}^T \mathbf{M}_3^{(t)} + \lambda_0 \mathbf{1}_{H\times W}^T \mathbf{B}^{(t+1)} + \mu \mathbf{V}^{(t)}}{\mathbf{C}^{(t)} \left(\mathbf{M}_3^{(t)}\right)^T \mathbf{M}_3^{(t)} + \lambda_0 \mathbf{C}^{(t)} \left(\mathbf{B}^{(t+1)}\right)^T \mathbf{B}^{(t+1)} + \mu \mathbf{C}^{(t)}}$.

  $\mathbf{U}^{(t+1)} \leftarrow \mathbf{U}^{(t)} .* \frac{\mu \mathbf{B}^{(t+1)} + \lambda_1 \mathbf{W} \mathbf{U}^{(t)}}{\mu \mathbf{U}^{(t)} + \lambda_1 \mathbf{D} \mathbf{U}^{(t)}}$.

  $\mathbf{V}^{(t+1)} \leftarrow \mathbf{V}^{(t)} .* \frac{\mu \mathbf{C}^{(t+1)} + \lambda_1 \mathbf{W} \mathbf{V}^{(t)}}{\mu \mathbf{V}^{(t)} + \lambda_1 \mathbf{D} \mathbf{V}^{(t)}}$.

  $\mathcal{G}_{Lk}^{(t+1)} \leftarrow \mathcal{G}_{Lk}^{(t)} .* \operatorname{diag}\left(\frac{\mathbf{P}^T \left(\mathbf{M}_4^{(t+1)}\right)^T \mathbf{X}_{(3)} \mathbf{C}^{(t+1)} \mathbf{1}_{Lk}}{\mathbf{P}^T \left(\mathbf{M}_4^{(t+1)}\right)^T \mathbf{M}_4^{(t+1)} \mathcal{G}_{Lk}^{(t)} \left(\mathbf{C}^{(t+1)}\right)^T \mathbf{C}^{(t+1)} \mathbf{P} \mathcal{G}_{Lk}^{(t)} \mathbf{1}_{Lk}} \right)$.

  $\mathcal{G}_L^{(t+1)} \leftarrow \frac{1}{k} \sum_{i=1}^{L} \left(u_i u_i^T \cdot \operatorname{tr} \left((u_i \otimes \mathbf{1}_k)^T \mathcal{G}_{Lk}^{(t+1)} (u_i \otimes \mathbf{1}_k) \right) \right)$.

  Break if stopping criteria are met.

  $t++$.

end

$\mathbf{A} \leftarrow \mathbf{B}^{(t+1)} \mathbf{C}^{(t+1)^T}$.

IV. Experiments and Results

In this section, we conduct experiments on one synthetic dataset and three real-world datasets. To further validate the performance of LA-NTF, we select several representative algorithms for comparison, including MVCNMF [31], GLNMF [29], graphL [57], gtvMBO [32], MV-NTF [45], EIC-NTF [47], MAC-U [58], GLMM [15], and MOGNMF [33]. Specifically, MVCNMF, GLNMF, graphL, gtvMBO, and MOGNMF are NMF-based unmixing methods; MV-NTF and EIC-NTF are NTF-based methods; MAC-U is an autoencoder-based unmixing method; and GLMM is a spectral variability-aware extension of the LMM that introduces band-dependent scaling factors to model endmember variability.

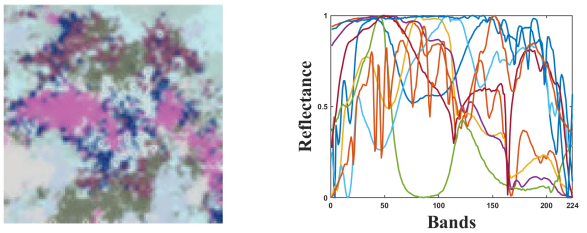
All algorithms are implemented in MATLAB R2022b on a computer equipped with an Intel Core i9-14900K CPU and 128 GB of RAM. The hyperparameters and initialization strategies of all competing algorithms are configured in accordance with their respective original publications.

A. Evaluation Metrics

Two widely adopted evaluation metrics are employed to quantitatively assess unmixing performance, namely spectral

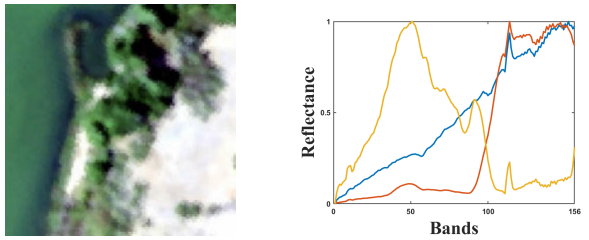
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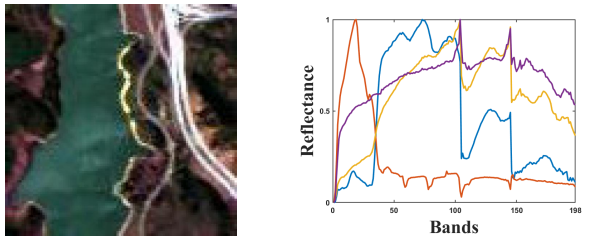
(a) (b)

Fig. 3. Synthetic Dataset. (a) Simulated image. (b) Endmembers spectra.



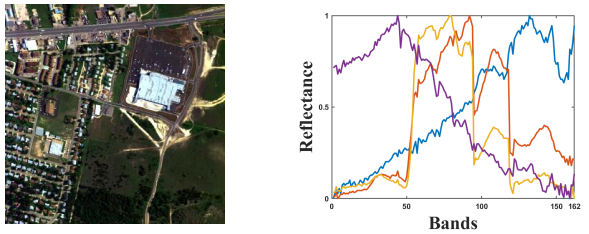
(a) (b)

Fig. 4. Samson Dataset. (a) RGB image. (b) Endmembers spectra.



(a) (b)

Fig. 5. Jasper Ridge Dataset. (a) RGB image. (b) Endmembers spectra.



(a) (b)

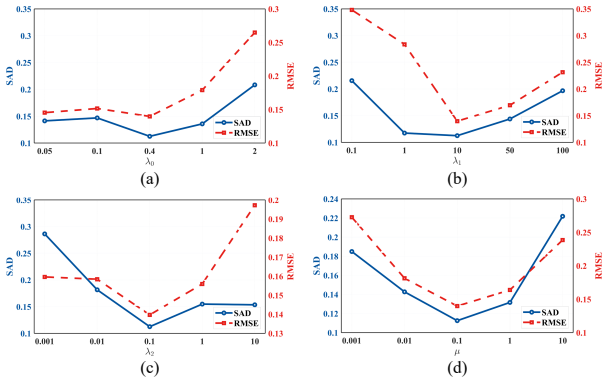
Fig. 6. Urban Dataset. (a) RGB image. (b) Endmembers spectra.

angle distance (SAD) and root-mean-square error (RMSE). Their mathematical definitions are given by

$$\mathrm{SAD}(\hat{\mathbf{e}}_j, \mathbf{e}_j) = \arccos\left(\frac{\hat{\mathbf{e}}_j^T \cdot \mathbf{e}_j}{\|\hat{\mathbf{e}}_j\| \cdot \|\mathbf{e}_j\|}\right), \tag{47}$$

$$\mathrm{RMSE}(\hat{\mathbf{A}}_j, \mathbf{A}_j) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{A}}_{ji} - \mathbf{A}_{ji}\|_2^2}, \tag{48}$$

where $\hat{\mathbf{e}}_j$ is the jth extracted endmember, and $\mathbf{e}_j$ denotes the true endmember. $\hat{\mathbf{A}}_j$ and $\mathbf{A}_j$ represent the estimated abundance of jth endmember and the corresponding real abundance.



(a) (b) (c) (d)

Fig. 7. Parameter sensitivity analysis of LA-NTF on the Urban dataset. (a) λ_0. (b) λ_1. (c) λ_2. (d) μ.

B. Hyperspectral Datasets

The proposed method is evaluated on one synthetic dataset and three real datasets. These datasets have different spatial dimensions and numbers of spectral bands, and the three real datasets were acquired by different sensors.

1) Synthetic Dataset: The synthetic dataset (SD) consists of 9 endmembers with 224 spectral bands, randomly selected from the USGS spectral library [59]. The abundance maps, each containing 100×100 pixels, are generated following the procedure described in [60], subject to the ANC and ASC. The simulated image and the spectral signatures of the nine endmembers are illustrated in Fig. 3. To evaluate robustness, zero-mean white Gaussian noise is added at SNRs of 10, 20, and 30 dB. No additional spectral variability is explicitly injected into the synthetic dataset; therefore, this dataset is mainly used to evaluate the basic unmixing ability and noise robustness of the proposed method under controlled noise levels.

2) Samson Dataset: The Samson dataset was acquired by the SAMSON sensor [61], comprising 952×952 pixels and 156 spectral bands spanning a wavelength range of 401-889 nm. As shown in Fig. 4, a subimage of 95×95 pixels is extracted for this experiment. Three dominant endmembers are considered: #1 Soil, #2 Tree, and #3 Water.

3) Jasper Ridge Dataset: The Jasper Ridge dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), comprising 512×614 pixels with 224 spectral bands spanning a wavelength range of 380-2500 nm. As shown in Fig. 5, a subimage of 100×100 pixels is extracted for this experiment. After removing bands affected by water vapor absorption and atmospheric interference, 198 bands are retained. Four dominant endmembers are identified in this dataset: #1 Soil, #2 Tree, #3 Water, and #4 Road.

4) Urban Dataset: The Urban dataset was acquired by the HYDICE sensor, comprising 307×307 pixels and 210 bands with spectral coverage from 400 to 2500 nm. After removing low-SNR bands and the water vapor absorption bands, 162 bands are retained. The RGB image and the endmember spectral signatures of the Urban dataset are shown in Fig. 6. Four dominant endmembers are considered: #1 Asphalt, #2 Grass, #3 Tree, and #4 Roof.

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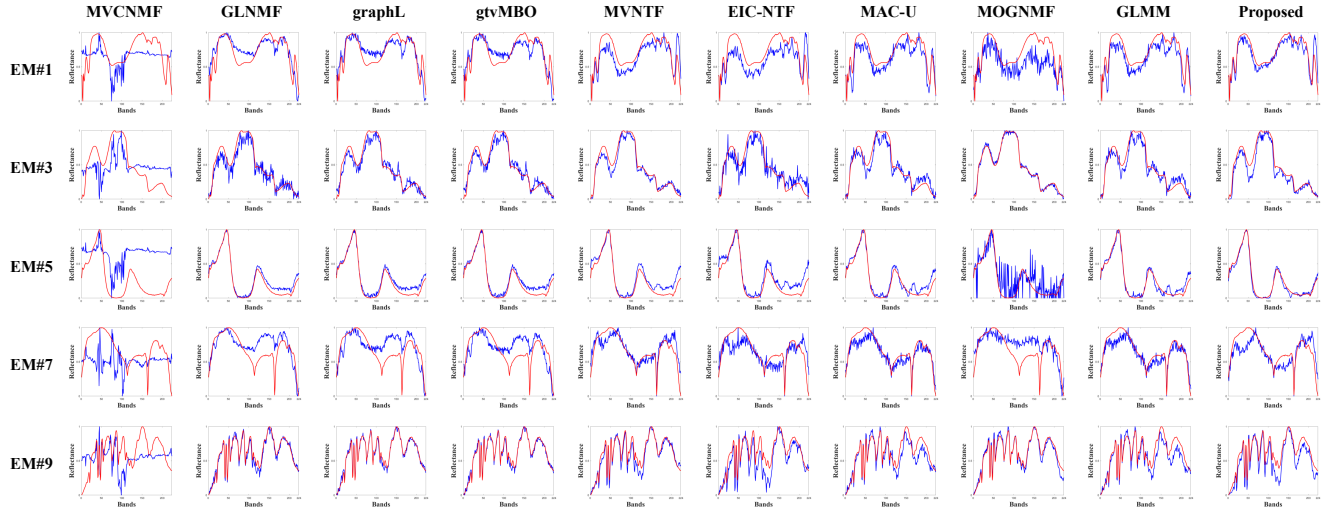


Fig. 8. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on SD at SNR=10dB.

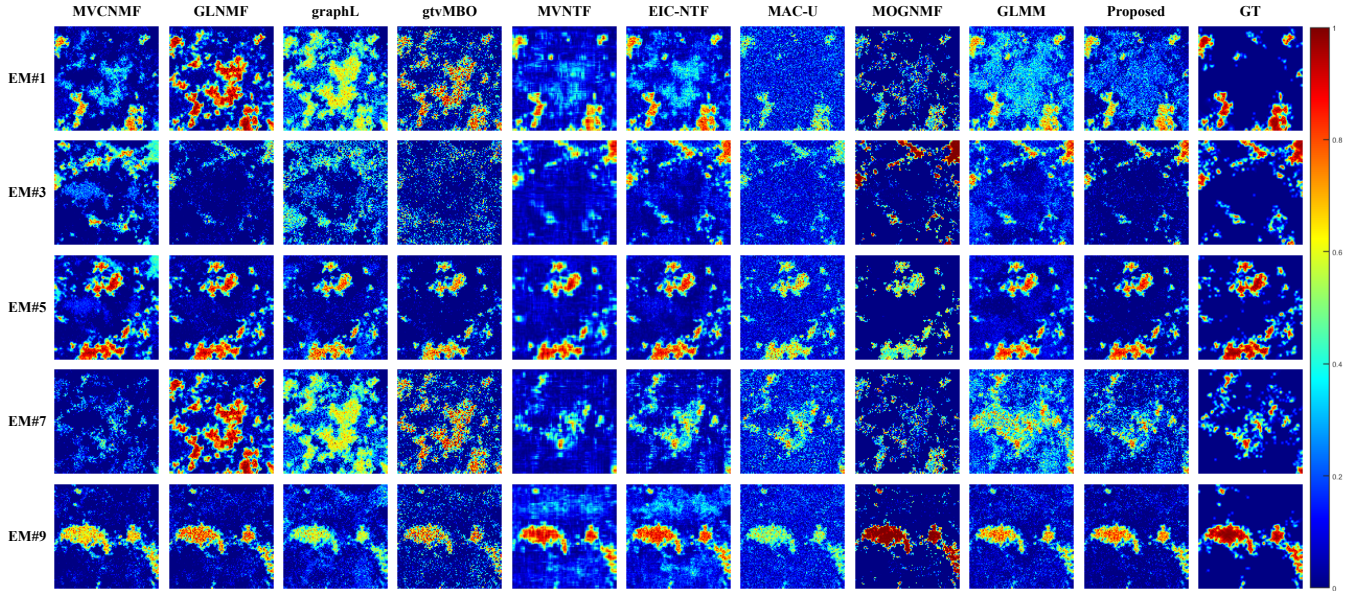


Fig. 9. Estimated abundance maps for different endmembers on SD at SNR=10dB.

TABLE II
PARAMETER SETTINGS IN EXPERIMENTS

Parameter	λ_0	λ_1	λ_2	μ
Synthetic data	0.4	10	0.1	0.1
Samson	0.4	0.001	50	0.001
Jasper	0.4	0.5	0.5	0.1
Urban	0.4	10	0.1	0.1

C. Parameter Setting

Since LA-NTF incorporates multiple constraint terms, it involves several regularization and penalty parameters. The hyperparameter settings for both the synthetic and real-world datasets are summarized in Table II. To further evaluate the influence of these parameters, we conduct a sensitivity analysis of λ_0, λ_1, λ_2, and μ on the Urban dataset, as visualized in Fig. 7. In addition, following the analyses of Qian et al. [45] and Xiong et al. [46], the BTD rank L is set to $L = 2/3 \max(H, W)$.

For the proposed LA-NTF, the endmember matrix $\mathbf{E}$ and the core tensor $\mathcal{G}_L$ are initialized randomly with values drawn from a uniform distribution over $[0, 1]$ across all datasets. The initialization of the abundance matrix $\mathbf{A}$ is critical, as it is decomposed into two matrices, $\mathbf{B}$ and $\mathbf{C}$, which are iteratively updated during optimization. Proper initialization of $\mathbf{A}$ not only accelerates the convergence of $\mathbf{B}$ and $\mathbf{C}$ but also helps preserve the physical interpretability of these factors throughout the optimization process. To this end, $\mathbf{A}$ is first initialized via the FCLS algorithm, ensuring that the initial abundance estimates satisfy both the ANC and ASC. The FCLS result is then decomposed using Singular Value Decomposition (SVD) to obtain the initial values of $\mathbf{B}$ and $\mathbf{C}$, followed by a projection onto the nonnegative orthant to

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TABLE III
QUANTITATIVE COMPARISONS ON THE SYNTHETIC DATASET AT SNR=10DB.

Method		MVCNMF	GLNMF	graphL	gtvMBO	MV-NTF	EIC-NTF	MAC-U	MOGNMF	GLMM	LA-NTF
SAD	EM#1	0.2951	0.1803	0.1653	0.1669	0.1742	0.1984	**0.1218**	0.1923	**0.1222**	0.1267
	EM#2	0.4918	0.4211	0.4022	0.3919	0.1079	0.2538	0.0982	**0.0525**	0.0962	**0.0824**
	EM#3	0.4665	0.1413	0.1622	0.1618	**0.0889**	0.2044	0.1567	**0.0457**	0.1531	0.1230
	EM#4	0.5860	0.0873	0.0995	0.1004	0.1168	0.3250	0.0952	**0.0478**	0.0922	**0.0761**
	EM#5	0.7197	**0.1272**	0.1438	0.1460	0.1894	0.2318	0.1696	0.4159	0.1423	**0.0856**
	EM#6	0.3614	**0.0378**	0.0720	0.0714	0.0872	0.1232	0.1068	0.0697	0.1077	**0.0623**
	EM#7	0.3539	0.5795	0.4175	0.4149	0.1598	0.2089	**0.1002**	0.2320	**0.1042**	0.2051
	EM#8	0.2563	0.4540	0.3947	0.3938	0.0783	0.1071	**0.0769**	0.0900	**0.0780**	0.1122
	EM#9	0.3894	0.0705	**0.0530**	**0.0545**	0.0989	0.1759	0.1365	0.0790	0.1396	0.1467
mean SAD		0.4356	0.2332	0.2122	0.2113	0.1224	0.2032	0.1180	0.1361	**0.1151**	**0.1133**
RMSE	EM#1	**0.1529**	0.3179	0.2864	0.2986	**0.1552**	0.1596	0.1929	0.2154	0.1914	0.1582
	EM#2	0.1259	0.2472	0.2244	0.2417	0.1340	0.1540	0.1631	**0.1136**	0.1478	**0.1079**
	EM#3	0.1742	0.1510	0.2039	0.2050	**0.1102**	0.1144	0.1606	**0.1122**	0.1240	0.1136
	EM#4	**0.1194**	0.2398	0.2103	0.2069	0.1525	0.1710	0.1767	0.1196	0.1587	**0.1137**
	EM#5	0.1038	**0.0521**	0.1085	0.1090	0.1100	0.1063	0.1606	0.1453	0.1003	**0.0854**
	EM#6	0.1503	**0.0735**	0.1524	0.1400	0.1081	0.1142	0.1600	0.1047	0.1303	**0.0841**
	EM#7	0.1472	0.2045	0.1743	0.1861	**0.1066**	**0.1099**	0.1274	0.2092	0.1369	0.1118
	EM#8	**0.1232**	0.2269	0.2202	0.2211	0.1307	0.1351	0.1587	0.2326	0.1605	**0.1301**
	EM#9	0.1037	**0.1025**	0.1717	0.1637	0.1432	0.1544	0.1836	**0.0977**	0.1988	0.1324
mean RMSE		0.1334	0.1795	0.1947	0.1969	**0.1278**	0.1354	0.1648	0.1500	0.1499	**0.1152**

TABLE IV
QUANTITATIVE RESULTS OF MEAN RMSE AND MEAN SAD FOR THE SYNTHETIC DATASET AT DIFFERENT SNRS.

Method		MVCNMF	GLNMF	graphL	gtvMBO	MV-NTF	EIC-NTF	MAC-U	MOGNMF	GLMM	LA-NTF
mean SAD	10dB	0.4356	0.2332	0.2122	0.2113	0.1224	0.2032	0.1180	0.1361	**0.1151**	**0.1133**
	20dB	0.0494	0.0706	0.0831	0.0815	0.0608	0.0638	0.0544	**0.0493**	0.0559	**0.0351**
	30dB	0.0527	**0.0088**	0.0232	0.0249	0.0522	0.0266	0.0230	0.0190	0.0241	**0.0075**
mean RMSE	10dB	0.1334	0.1795	0.1947	0.1969	**0.1278**	0.1354	0.1648	0.1500	0.1499	**0.1152**
	20dB	0.0624	0.0645	0.1284	0.0794	0.0829	0.0686	0.0905	**0.0624**	0.0650	**0.0622**
	30dB	0.0341	0.0273	0.0789	**0.0178**	0.0706	0.0362	0.0412	0.0279	0.0324	**0.0266**

satisfy the nonnegativity constraints before the multiplicative updates.

D. Experiments on Synthetic Dataset and Analysis

1) Unmixing Performance: On the SD dataset with an SNR of 10 dB, the quantitative results in terms of RMSE and SAD for each endmember are presented in Table III, where the best and second-best results are highlighted in red and blue, respectively. The extracted endmembers and their corresponding abundance maps are visualized in Fig. 8 and Fig. 9. Since the SD dataset contains a large number of endmembers, we select EM#1, #3, #5, #7, and #9 for display. As shown, LA-NTF achieves the best performance in terms of both SAD and RMSE. The visualization results indicate that LA-NTF is less susceptible to noise interference, demonstrating superior performance in both endmember extraction and abundance estimation. This suggests that the joint spatial–spectral modeling enabled by the tensor structure, together with the TV and MV regularization terms, effectively suppresses noise interference while maintaining robust spectral feature extraction and abundance inversion performance under high noise conditions.

2) Noise Robustness Analysis: To evaluate the robustness of LA-NTF against noise, we further introduced zero-mean white Gaussian noise with SNRs of 20 dB and 30 dB to the synthetic dataset. Table IV presents the quantitative results for all competing algorithms across the three noise levels. As illustrated in Fig. 10, LA-NTF achieves superior performance in most cases, except for a second-best RMSE result at SNR = 30 dB. When the noise level is relatively low, gtvMBO benefits from its graph TV operator to accurately extract spatial features. In contrast, LA-NTF maintains stable performance under high-noise conditions, which can be attributed to the joint spatial–spectral regularization enforced by the tensor factorization framework, together with the incorporated TV and MV constraints. These components jointly regularize the solution space, ensuring algorithmic stability and robustness across varying noise intensities.

E. Experiments on Real Datasets

1) Samson Dataset: The quantitative results are summarized in Table V, while the visualization of the extracted endmembers and estimated abundance maps is provided in Fig. 11 and 12, respectively. The Samson dataset represents a

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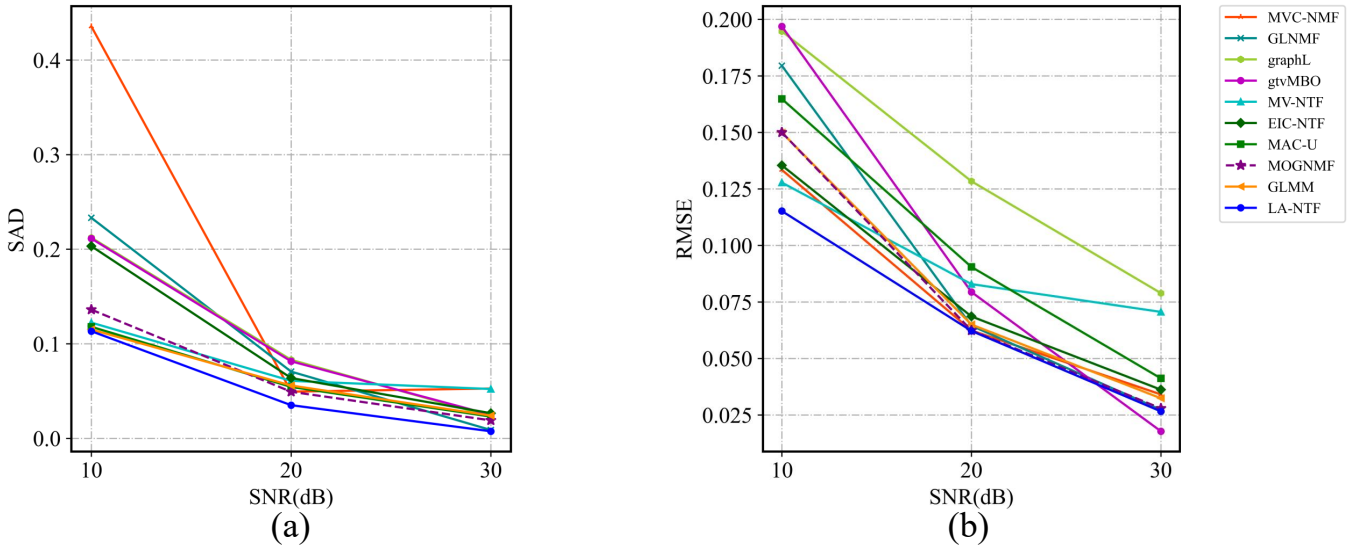


Fig. 10. Performance comparison of different algorithms under various SNRs on the synthetic dataset. (a) SAD. (b) RMSE.

TABLE V
QUANTITATIVE RESULTS OF MEAN RMSE AND MEAN SAD FOR SAMSON DATASET.

Method		MVCNMF	GLNMF	graphL	gtvMBO	MV-NTF	EIC-NTF	MAC-U	MOGNMF	GLMM	LA-NTF
SAD	Soil	**0.0305**	**0.0289**	0.0343	0.0357	0.2820	0.1222	0.0460	0.1254	0.1766	0.1221
	Tree	0.0400	**0.0330**	0.0491	0.0727	0.0796	0.0411	0.0334	0.0579	0.0528	**0.0273**
	Water	0.3589	0.3426	0.2245	0.3639	**0.1201**	0.1688	**0.1193**	0.2653	0.3284	0.1493
mean SAD		0.1431	0.1348	0.1026	0.1574	0.1606	0.1107	**0.0662**	0.1495	0.1859	**0.0996**
RMSE	Soil	0.2233	0.2045	**0.1310**	0.1363	0.3724	0.1870	0.2781	0.1640	0.2145	**0.1111**
	Tree	0.2432	0.2012	**0.1545**	0.1685	0.2716	0.2134	0.1678	0.1603	0.1684	**0.1086**
	Water	0.4181	0.3317	**0.1670**	0.1960	0.4582	0.3252	0.3231	0.2180	0.3026	**0.1225**
mean RMSE		0.2949	0.2458	**0.1508**	0.1669	0.3674	0.2419	0.2563	0.1808	0.2285	**0.1141**

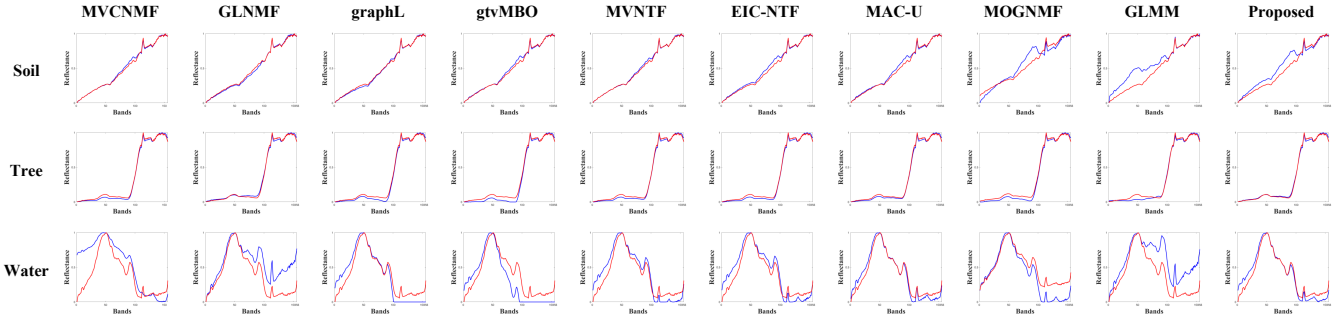


Fig. 11. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on Samson dataset.

relatively straightforward unmixing scenario as it contains only three endmembers with moderate inter-class coupling. Consequently, most competing algorithms demonstrate satisfactory performance. Notably, MAC-U, which leverages the powerful data-fitting capability of deep learning networks through an autoencoder architecture and Multilayer Perceptrons (MLPs), achieves the lowest mean SAD.

Meanwhile, LA-NTF achieves the second-lowest mean SAD and the lowest mean RMSE, demonstrating competitive performance in both endmember and abundance estimation. A closer inspection of the abundance maps reveals that LA-NTF produces sharper boundaries and more coherent spatial structures compared to other methods. This superior visual quality can be attributed to the inherent advantage of tensor-based factorization, which treats the HSI as a third-order tensor rather than a flattened matrix. By avoiding vectorization, LA-NTF effectively preserves the intrinsic spatial-spectral correlations, thereby enhancing the robustness of abundance inversion and reducing estimation artifacts.

Nevertheless, LA-NTF yields a higher SAD for the Soil endmember than GLNMF and MVCNMF, although it achieves the lowest abundance RMSE for all three endmembers. This

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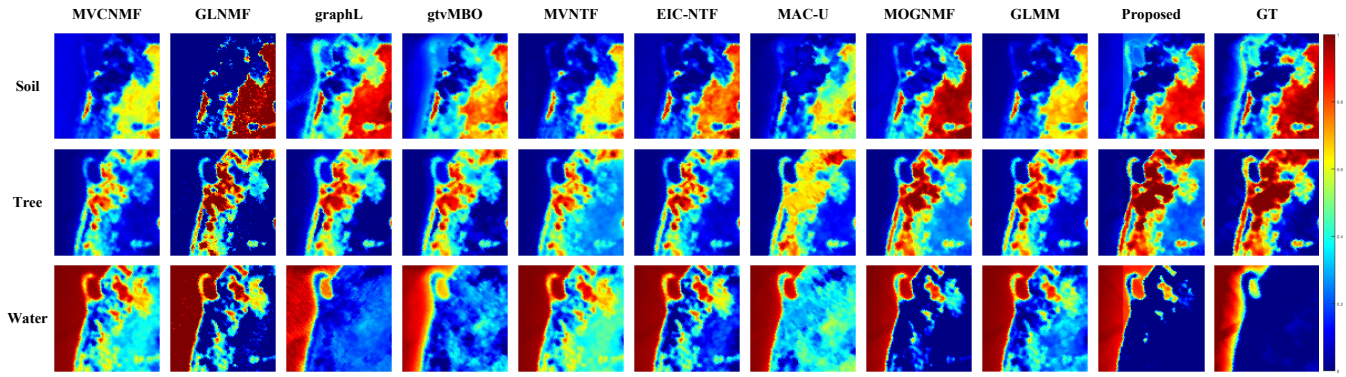


Fig. 12. Estimated abundance maps for different endmembers on Samson dataset.

TABLE VI
QUANTITATIVE RESULTS OF MEAN RMSE AND MEAN SAD FOR JASPER RIDGE DATASET

Method		MVCNMF	GLNMF	graphL	gtvMBO	MV-NTF	EIC-NTF	MAC-U	MOGNMF	GLMM	LA-NTF
SAD	Soil	0.1138	0.1098	**0.0800**	0.1439	**0.0836**	0.2725	0.1184	0.1068	0.1215	0.2050
	Tree	0.1585	**0.0983**	0.2170	0.2292	0.2202	0.2672	0.1805	**0.0561**	0.1552	0.1252
	Water	0.2382	**0.0628**	0.3918	0.3919	0.2959	0.1547	0.1487	0.1301	0.1456	**0.0617**
	Road	**0.0609**	0.7719	0.3686	0.1029	0.2171	0.0998	0.0825	**0.0481**	0.1231	0.1347
mean SAD		0.1429	0.2607	0.2644	0.2170	0.2042	0.1986	0.1325	**0.0853**	0.1364	**0.1317**
RMSE	Soil	**0.1131**	0.2221	0.1599	0.1570	0.2869	0.2065	0.1405	0.1542	**0.1294**	0.1751
	Tree	0.1511	**0.0963**	0.1907	0.1942	0.2007	0.2395	0.1567	0.1570	0.1386	**0.1319**
	Water	0.1920	0.1163	0.1744	**0.1059**	0.2269	0.1347	0.1656	0.2801	0.1925	**0.1008**
	Road	**0.1108**	0.2645	0.1210	0.1389	0.1924	0.1922	0.1183	0.1593	**0.1055**	0.1470
mean RMSE		0.1418	0.1748	0.1615	0.1490	0.2267	0.1932	0.1453	0.1877	**0.1415**	**0.1387**

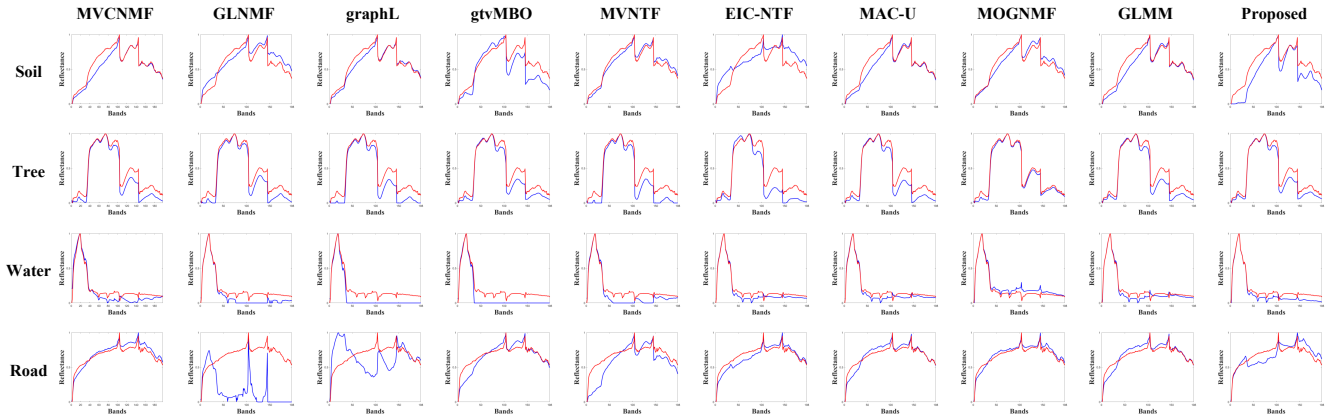


Fig. 13. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on Jasper Ridge dataset.

discrepancy indicates that more accurate abundance estimation does not necessarily translate into lower SAD for every endmember. The higher Soil SAD may reflect the trade-off between spectral reconstruction, spatial regularization, and loss-induced scaling in the joint optimization of LA-NTF. These components improve the overall unmixing results but do not directly minimize the spectral angle of each individual endmember.

2) Jasper Ridge Dataset: Quantitative results are presented in Table VI, and the visualization results of the extracted endmembers and estimated abundance maps are shown in Fig. 13 and 14, respectively.

As shown in Table VI, MOGNMF achieves the lowest mean SAD of 0.0853, while LA-NTF obtains the second-lowest mean SAD of 0.1317 and the lowest mean RMSE of 0.1387. These results indicate that MOGNMF provides more accurate endmember spectral estimates on average, whereas LA-NTF achieves more accurate abundance estimates.

For the Road endmember, MOGNMF achieves the lowest SAD of 0.0481, followed by MVCNMF with 0.0609, whereas LA-NTF yields a higher SAD of 0.1347. MOGNMF may benefit from its adaptive multi-order graph regularization, which exploits relationships among pixels to constrain the factorization. The relatively large error of LA-NTF may be as-

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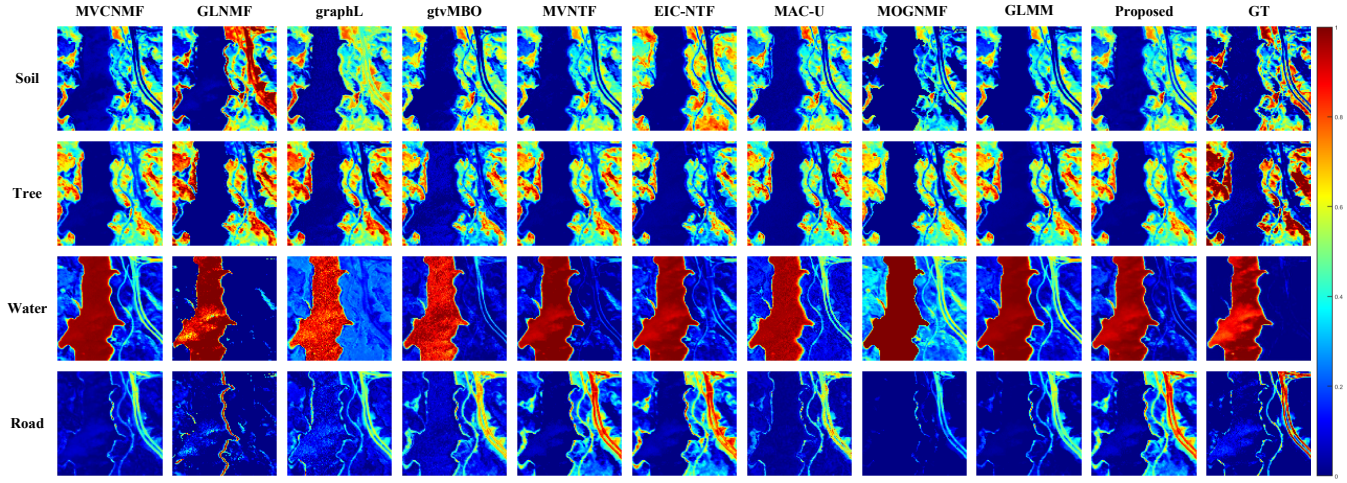


Fig. 14. Estimated abundance maps for different endmembers on Jasper Ridge dataset.

TABLE VII
QUANTITATIVE RESULTS OF MEAN RMSE AND MEAN SAD FOR URBAN DATASET

Method		MVCNMF	GLNMF	graphL	gtvMBO	MV-NTF	EIC-NTF	MAC-U	MOGNMF	GLMM	LA-NTF
SAD	Asphalt	0.2588	0.3238	0.2380	0.2455	0.2154	0.2016	0.2527	0.1993	**0.1824**	**0.1273**
	Grass	0.6452	0.1979	0.1609	0.1543	0.1474	0.4226	0.0932	0.1396	**0.0894**	**0.0509**
	Tree	0.5503	0.1590	**0.0420**	**0.0418**	0.1643	0.0871	0.3372	0.2589	0.1697	0.1042
	Roof	0.2090	0.6306	0.5299	0.5433	0.5704	**0.1778**	0.4235	0.1906	0.4187	**0.1674**
mean SAD		0.4158	0.3278	0.2427	0.2462	0.2744	0.2223	0.2767	**0.1971**	0.2151	**0.1125**
RMSE	Asphalt	0.3014	0.3228	**0.1880**	0.1924	0.2179	0.3944	0.2872	0.4051	0.2112	**0.1488**
	Grass	0.2686	0.1942	**0.1234**	**0.1270**	0.2327	0.3143	0.4047	0.7036	0.1608	0.1505
	Tree	0.2958	**0.1316**	0.1680	0.1698	0.2105	0.2586	0.3199	0.4446	**0.1259**	0.1465
	Roof	0.1878	0.1915	**0.1246**	0.1295	0.1502	0.1895	0.2180	0.2413	0.1446	**0.1128**
mean RMSE		0.2634	0.2100	**0.1510**	0.1547	0.2028	0.2892	0.3075	0.4487	0.1606	**0.1397**

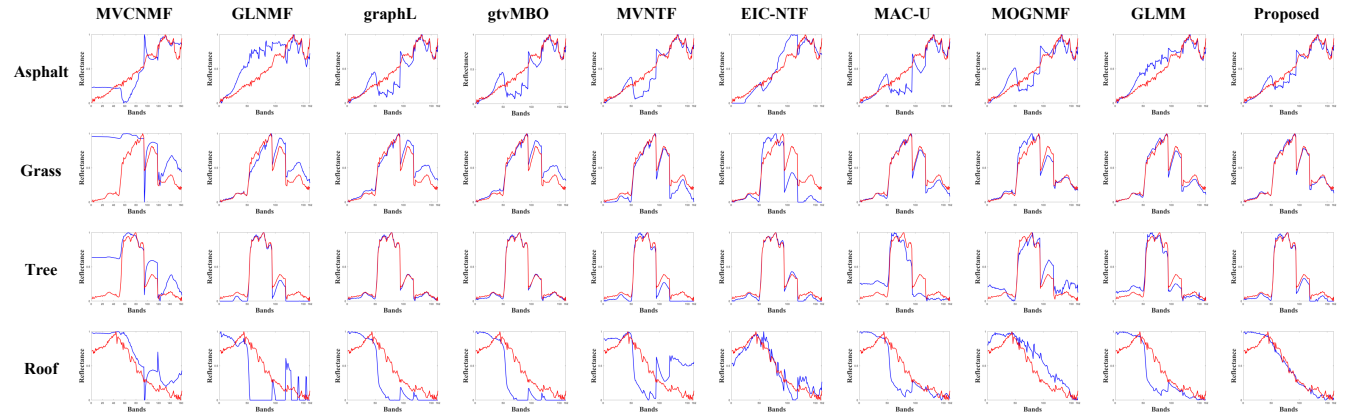


Fig. 15. Estimated endmembers by different methods (blue curves) and the corresponding ground truth (red curves) on Urban dataset.

sociated with spectral similarity between Road and Soil, which makes their contributions more difficult to distinguish. In addition, if road regions form narrow spatial structures, spatial smoothness regularization may weaken their separation from adjacent materials, potentially affecting both the estimated abundances and the corresponding endmember spectrum.

For abundance estimation, LA-NTF achieves the lowest mean RMSE, including the lowest RMSE for Water and the second-lowest RMSE for Tree. MVCNMF and GLMM obtain the lowest RMSE for Soil and Road, respectively.

3) Urban Dataset: The quantitative results are presented in Table VII, and the visualization results of the extracted endmembers and estimated abundance maps are shown in Fig. 15 and 16, respectively. The Urban dataset contains four endmembers and represents a challenging scenario with complex spectral variability.

As shown in Table VII, LA-NTF achieves the best mean SAD and RMSE, demonstrating its superior overall unmixing

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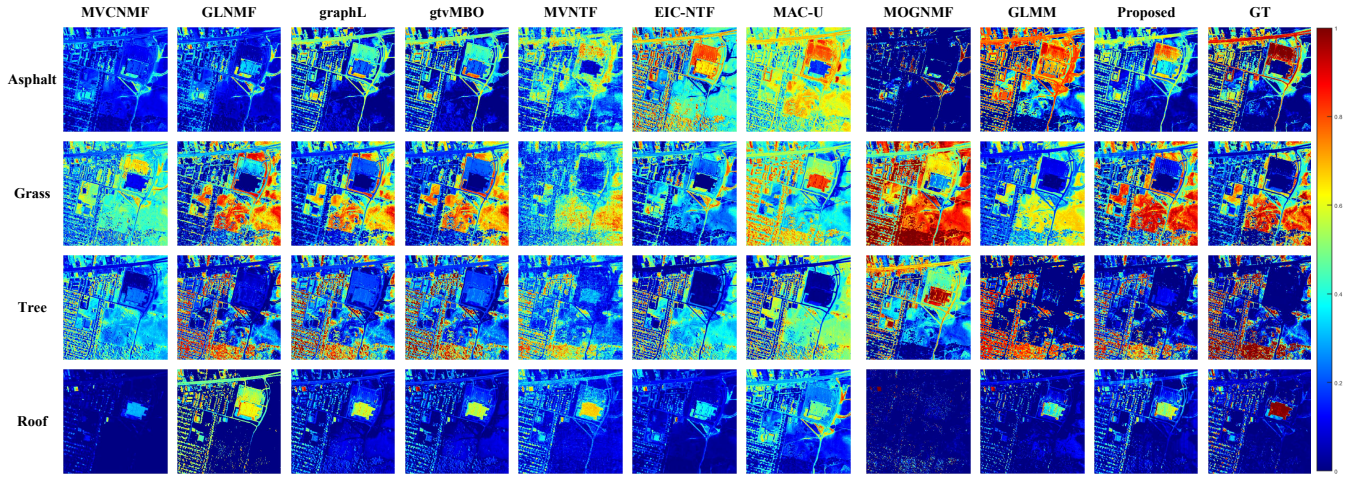


Fig. 16. Estimated abundance maps for different endmembers on Urban dataset.

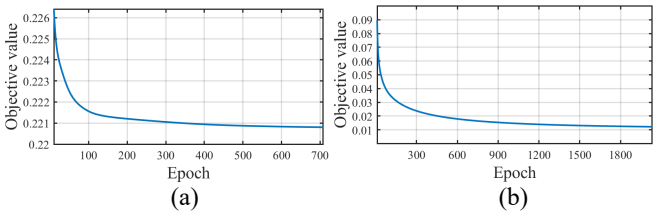


Fig. 17. Convergence curves on the real datasets. (a) Jasper Ridge. (b) Urban.

capability. GLMM also obtains competitive performance due to its ability to characterize spectral variability through band-wise scaling factors. Compared with these methods, LA-NTF further benefits from the LALMM-based variability representation and tensor-based spatial-spectral modeling.

For endmember extraction, LA-NTF achieves the best SAD values for Asphalt, Grass, and Roof, while graph-based methods obtain better performance for the Tree endmember. Specifically, graphL and gtvMBO achieve lower SAD values for Tree, which can be attributed to the effectiveness of graph-based spatial regularization in exploiting the spatial continuity of this endmember. In contrast, the advantage of LALMM-based spectral variability modeling is less significant for Tree, where the spectral variability contribution is relatively limited.

For abundance estimation, LA-NTF achieves the best overall RMSE and produces abundance maps with clearer boundaries and more coherent spatial structures. This demonstrates that the tensor-based factorization with TV regularization effectively preserves the spatial-spectral correlations of the HSI and improves abundance inversion accuracy.

F. Spectral Variability Analysis

To further investigate the effectiveness of the proposed LALMM and LA-NTF in addressing spectral variability, we visualize the extracted endmember bundles and the statistical distribution of the core tensor $\mathcal{G}_L$. Fig. 18 and 19 illustrate the spectral variability results on the Jasper Ridge and Urban datasets, respectively. In subfigures (a)-(d), the red curves represent the ground truth endmembers, the dark blue curves denote the estimated base endmembers, and the dense light blue curves represent the variant endmembers reconstructed by incorporating the core tensor $\mathcal{G}_L$.

As observed in the spectral plots, the generated light-blue bundles generally encompass the ground-truth signatures, providing empirical evidence that LA-NTF can capture intra-class spectral variability in the considered scenes. Furthermore, the histograms in subfigure (e) illustrate the distributions of the learned latent-domain coefficients in $\mathcal{G}_L$. It should be emphasized that the entries of $\mathcal{G}_L$ are not direct estimates of the original pixel-wise scaling factors in Γ, nor of the Markov-chain probabilities p_i and q_i. Instead, $\mathcal{G}_L$ operates jointly with the spatial basis matrices $\mathbf{B}_i$ and $\mathbf{C}_i$ to provide a low-rank representation of the scaling effect of Γ in the original spatial domain. Therefore, the physical range of the original scaling factors does not need to be imposed element-wise on the latent coefficients of $\mathcal{G}_L$, and individual entries of $\mathcal{G}_L$ may exceed one without contradicting the attenuation range derived for Γ.

For the Jasper Ridge dataset, the elements of $\mathcal{G}_L$ in Fig. 18(e) are tightly concentrated around 1.0, indicating that only a relatively small deviation from the identity-core configuration is required to represent the spatial scaling variability in this scene. In contrast, the distribution for the Urban dataset in Fig. 19(e) is considerably broader, ranging from 0 to approximately 1.5. This broader distribution indicates that LA-NTF employs more heterogeneous latent-domain modulation to characterize the more complex spectral variations in the Urban scene. Together with the reconstructed endmember bundles, these results demonstrate that the physics-motivated core-tensor representation provides a flexible mechanism for modeling different levels of spectral variability.

G. Convergence Analysis

To evaluate the convergence behavior of LA-NTF, we record the objective function value with respect to the number of iterations on the Jasper Ridge and Urban datasets, as shown in Fig. 17.

The objective function values decrease rapidly in the early iterations and then gradually become stable on both datasets,

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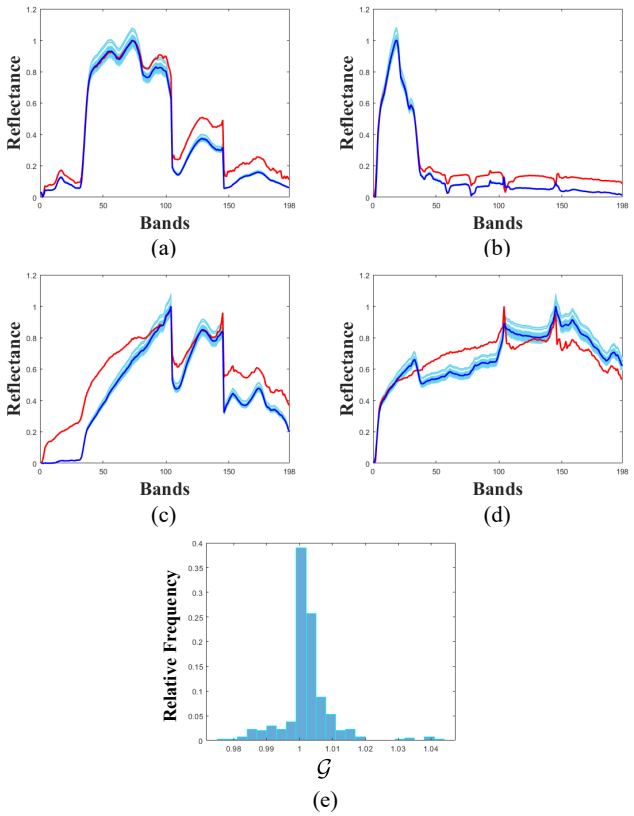


Fig. 18. Visualization of spectral variability on Jasper Ridge dataset. (a)-(d) Extracted endmember bundles for the four endmembers. (e) Histogram illustrating the relative frequency distribution of the elements in $\mathcal{G}_L$.

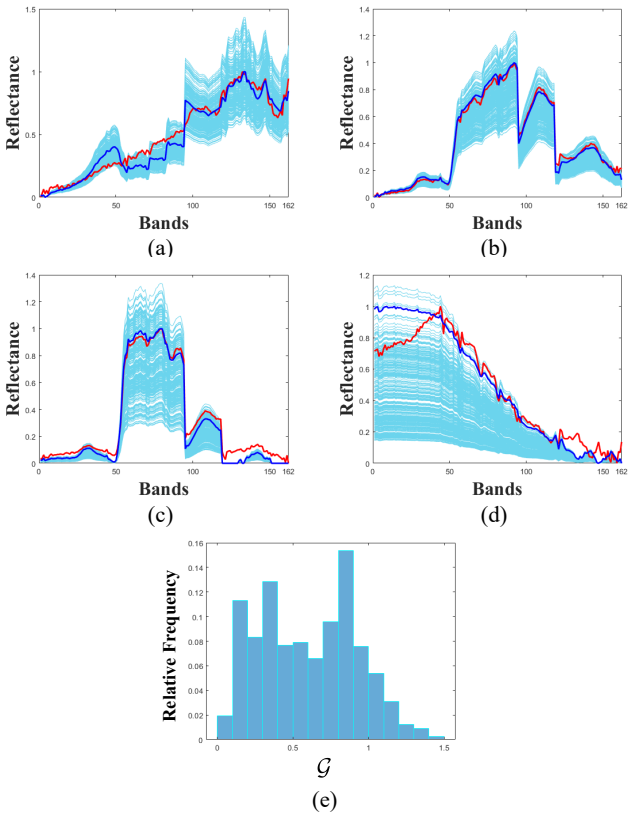


Fig. 19. Visualization of spectral variability on Urban dataset. (a)-(d) Extracted endmember bundles for the four endmembers. (e) Histogram illustrating the relative frequency distribution of the elements in $\mathcal{G}_L$.

TABLE VIII
ABLATION EXPERIMENTS ON JASPER RIDGE AND URBAN DATASET

Configurations				Jasper Ridge		Urban	
Model	$\mathcal{G}_L$	MV	TV	SAD	RMSE	SAD	RMSE
LALMM	✓	✗	✓	0.1971	0.1513	0.1764	0.1566
LALMM	✓	✓	✗	0.1810	0.1682	0.1529	0.1753
BTD	$\mathbf{I}_L$	✓	✓	0.1437	0.1556	0.1381	0.1624
LALMM	✓	✗	✗	0.1987	0.1789	0.2043	0.2531
LMM	-	✓	✓	0.2030	0.1728	0.3263	0.2786
LALMM	✓	✓	✓	**0.1317**	**0.1387**	**0.1125**	**0.1397**

indicating that the proposed multiplicative update scheme exhibits stable empirical convergence behavior. Although Jasper Ridge and Urban contain complex material distributions and strong spectral variability, LA-NTF can still steadily optimize the objective function and converge to stable solutions. Overall, these results provide empirical evidence that LA-NTF exhibits stable convergence behavior on the real datasets.

H. Ablation Experiments

To further validate the effectiveness of LALMM and LA-NTF, we conduct ablation analyses on the Jasper Ridge and Urban datasets. Specifically, we investigate the contributions of the core tensor $\mathcal{G}_L$, the MV regularization term, and the TV regularization term within the LA-NTF framework. Moreover, an LMM+MV+TV configuration is introduced to evaluate the contribution of the proposed spectral variability modeling by removing the LALMM-based tensor formulation while retaining the same regularization terms. The quantitative results are presented in Table VIII, with the best results highlighted in bold. The complete LA-NTF model achieves the best performance on both datasets, demonstrating the effectiveness of the proposed components and model formulation.

Upon removing the MV term, SAD degrades significantly on both datasets, as the reduced constraint on endmembers makes it difficult for the algorithm to identify better solutions in the enlarged solution space. The RMSE of abundance estimation also increases accordingly, which can be attributed to the iterative optimization process and the inherent coupling between endmembers and abundances in the unmixing model. A similar trend is observed when the TV term is removed. When both regularization terms are removed simultaneously, the performance degradation becomes more severe, further demonstrating the importance of regularization in improving the optimization stability and estimation accuracy.

To investigate the contribution of the LALMM-based core tensor formulation, we replace the learnable core tensor $\mathcal{G}_L$ with an identity tensor $\mathbf{I}_L$, corresponding to the standard BTD formulation. The resulting performance degradation on both datasets demonstrates the importance of the proposed core tensor modeling. On the Jasper Ridge dataset, the degradation is relatively modest, which is consistent with the fact

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TABLE IX
COMPUTATIONAL COST OF DIFFERENT METHODS ON JASPER RIDGE DATASET

Method	MVCNMF	GLNMF	graphL	gtvMBO	MV-NTF	EIC-NTF	MAC-U	MOGNMF	GLMM	LA-NTF
Runtime (s)	52.49	3.86	0.10	1.37	9.10	48.65	76.41	50.49	79.80	66.51
Peak Memory (MB)	60.96	15.58	2.51	2.53	15.50	66.01	31.00	37.70	86.96	62.00
Iterations	150	1000	100	100	231	464	15	1000	100	708

that the elements of $\mathcal{G}_L$ are mainly distributed around 1.0 on this dataset. Nevertheless, the RMSE still increases by 0.0169, indicating that $\mathcal{G}_L$ provides meaningful guidance for optimizing the spatial bases $\mathbf{B}$ and $\mathbf{C}$ during iterative updates, even when its values are close to unity. On the Urban dataset, the degradation is more significant, with SAD increasing by 0.0256 and RMSE by 0.0227, which agrees with the broader distribution of $\mathcal{G}_L$ elements observed in the spectral variability analysis.

Furthermore, compared with the LMM+MV+TV variant, the proposed LA-NTF achieves better unmixing performance on both datasets. This improvement indicates that the performance gain does not merely originate from the MV and TV regularization terms, but also benefits from the LALMM-based spectral variability modeling and the corresponding tensor representation.

I. Computational Cost

To evaluate the computational cost of the proposed algorithm, we conduct computational cost experiments on the Jasper Ridge dataset, which has a moderate spatial size and endmember number among the three real datasets. All algorithms are implemented in MATLAB R2022b and run on a computer equipped with an Intel Core i9-14900K CPU and 128 GB of RAM. We assess the computational cost using runtime, peak memory usage, and the number of iterations, as reported in Table IX.

LA-NTF is built upon LALMM, where spectral variability is modeled through additional physical parameters, resulting in increased degrees of freedom and optimization complexity. Consequently, additional iterations are required, and regularization terms are incorporated to constrain the enlarged solution space and achieve a stable convergence point. Moreover, the alternating updates of multiple factor matrices inherent to NTF-based algorithms further increase the computational burden. Overall, the proposed method introduces additional computational costs while maintaining an acceptable runtime compared with other spectral variability modeling approaches.

V. CONCLUSION

In this article, we propose a novel physics-derived spectral mixing model termed the Loss-Aware LMM (LALMM) and develop an NTF-based hyperspectral unmixing algorithm (LA-NTF). We propose LALMM based on the absorbing Markov chain framework, and introduce a propagation loss q and a capture probability p to simulate the inevitable energy attenuation and propagation loss experienced during photon transmission. Specifically, within LA-NTF, the core tensor $\mathcal{G}_L$ in the $(L, L, 1)$-BTD framework is leveraged to capture spectral variability and scale variations. Furthermore, TV and MV constraints are incorporated into LA-NTF to ensure the physical validity of the solution and to regularize the solution space.

In future work, we will incorporate deep learning techniques to address the simplifications in the current formulation, particularly by replacing the shared $\mathcal{G}_L$ with a physics-guided data-driven module, such as an attention-based network, to adaptively estimate pixel-wise propagation coefficients while preserving the probabilistic interpretation of the Markov chain. Additionally, we will investigate deep unrolling frameworks to achieve tighter integration between the physical structure of LALMM and the unmixing algorithm, combining physical interpretability with data-driven optimization.

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APPENDIX
DERIVATION OF THE LIMITING TRANSITION MATRIX

In this appendix, we provide the detailed derivation of the limiting transition matrix $\lim_{n\to\infty} \mathbf{M}_i^n$ presented in (14).

It is a well-established property of absorbing Markov chains in [56] that as $n \to \infty$, the matrix $\mathbf{M}_i^n$ converges to:

$$\lim_{n\to\infty} \mathbf{M}_i^n = \begin{pmatrix} \mathbf{0}_{k\times k} & (\mathbf{I}_k - \mathbf{Q}_i)^{-1}\mathbf{R}_i \\ \mathbf{0}_{2\times k} & \mathbf{I}_2 \end{pmatrix}, \tag{49}$$

where $\mathbf{N}_i = (\mathbf{I}_k - \mathbf{Q}_i)^{-1}$ is known as the fundamental matrix of the Markov chain. To compute this limit, our primary task is to calculate $(\mathbf{I}_k - \mathbf{Q}_i)^{-1}\mathbf{R}_i$.

First, we rewrite $\mathbf{Q}_i$ and $\mathbf{R}_i$ from (13) into a more compact vector outer product form. Let $\mathbf{1}_k$ denote a $k\times 1$ column vector of ones. The matrix $\mathbf{Q}_i$ consists of identical rows, which can be expressed as:

$$\mathbf{Q}_i = (1 - p_i - q_i)\mathbf{1}_k\mathbf{a}_i^T. \tag{50}$$

Similarly, the matrix $\mathbf{R}_i$ can be expressed as:

$$\mathbf{R}_i = \mathbf{1}_k \begin{pmatrix} q_i & p_i \end{pmatrix}. \tag{51}$$

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To find the inverse of $(\mathbf{I}_k - \mathbf{Q}_i)$, we apply the Sherman-Morrison formula, which states that for an invertible matrix $\mathbf{A}$ and vectors $\mathbf{u}, \mathbf{v}$:

$$(\mathbf{A} - \mathbf{u}\mathbf{v}^T)^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^T\mathbf{A}^{-1}}{1 - \mathbf{v}^T\mathbf{A}^{-1}\mathbf{u}}. \tag{52}$$

By substituting $\mathbf{A} = \mathbf{I}_k$, $\mathbf{u} = (1 - p_i - q_i)\mathbf{1}_k$, and $\mathbf{v} = \mathbf{a}_i$, we evaluate the scalar term in the denominator $\mathbf{v}^T\mathbf{A}^{-1}\mathbf{u}$:

$$\begin{aligned} \mathbf{v}^T\mathbf{A}^{-1}\mathbf{u} &= \mathbf{a}_i^T\mathbf{I}_k(1 - p_i - q_i)\mathbf{1}_k \\ &= (1 - p_i - q_i)(\mathbf{a}_i^T\mathbf{1}_k). \end{aligned} \tag{53}$$

According to the ASC in (2), we have $\sum_{j=1}^{k} a_{i,j} = 1$, which is equivalent to $\mathbf{a}_i^T\mathbf{1}_k = 1$. Therefore:

$$\mathbf{v}^T\mathbf{A}^{-1}\mathbf{u} = 1 - p_i - q_i. \tag{54}$$

Applying this back to the Sherman-Morrison formula, the fundamental matrix becomes:

$$\begin{aligned} (\mathbf{I}_k - \mathbf{Q}_i)^{-1} &= \mathbf{I}_k + \frac{(1 - p_i - q_i)\mathbf{1}_k\mathbf{a}_i^T}{1 - (1 - p_i - q_i)} \\ &= \mathbf{I}_k + \frac{1}{p_i + q_i}\mathbf{Q}_i. \end{aligned} \tag{55}$$

Next, we multiply the fundamental matrix by $\mathbf{R}$ to obtain the final absorption probabilities:

$$\begin{aligned} (\mathbf{I}_k - \mathbf{Q}_i)^{-1}\mathbf{R}_i &= \left(\mathbf{I}_k + \frac{1}{p_i + q_i}\mathbf{Q}_i\right)\mathbf{R}_i \\ &= \mathbf{R}_i + \frac{1}{p_i + q_i}\mathbf{Q}_i\mathbf{R}_i. \end{aligned} \tag{56}$$

Notice that $\mathbf{Q}_i\mathbf{R}_i$ can be simplified using the outer product definitions and the ASC ($\mathbf{a}_i^T\mathbf{1}_k = 1$):

$$\begin{aligned} \mathbf{Q}_i\mathbf{R}_i &= \left((1 - p_i - q_i)\mathbf{1}_k\mathbf{a}_i^T\right)\begin{pmatrix}\mathbf{1}_k & \begin{pmatrix} q_i & p_i \end{pmatrix}\end{pmatrix} \\ &= (1 - p_i - q_i)\mathbf{1}_k(\mathbf{a}_i^T\mathbf{1}_k)\begin{pmatrix} q_i & p_i \end{pmatrix} \\ &= (1 - p_i - q_i)\mathbf{R}. \end{aligned} \tag{57}$$

Substituting this back yields:

$$\begin{aligned} (\mathbf{I}_k - \mathbf{Q}_i)^{-1}\mathbf{R}_i &= \mathbf{R}_i + \frac{1 - p_i - q_i}{p_i + q_i}\mathbf{R}_i \\ &= \frac{1}{p_i + q_i}\mathbf{R}_i. \end{aligned} \tag{58}$$

Thus, the limiting transition matrix is rigorously proven to be:

$$\lim_{n\to\infty} \mathbf{M}_i^n = \begin{pmatrix} \mathbf{0}_{k\times k} & \frac{1}{p_i+q_i}\mathbf{R}_i \\ \mathbf{0}_{2\times k} & \mathbf{I}_2 \end{pmatrix}. \tag{59}$$

This concludes the proof.

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